



Top 34 CAT Scheduling Questions With Video Solutions

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Questions

Instructions

A visa processing office (VPO) accepts visa applications in four categories - US, UK, Schengen, and Others. The applications are scheduled for processing in twenty 15- minute slots starting at 9:00 am and ending at 2:00 pm. Ten applications are scheduled in each slot.

There are ten counters in the office, four dedicated to US applications, and two each for UK applications, Schengen applications and Others applications. Applicants are called in for processing sequentially on a first-come-first-served basis whenever a counter gets freed for their category. The processing time for an application is the same within each category. But it may vary across the categories. Each US and UK application requires 10 minutes of processing time. Depending on the number of applications in a category and time required to process an application for that category, it is possible that an applicant for a slot may be processed later.

On a particular day, Ira, Vijay and Nandini were scheduled for Schengen visa processing in that order. They had a 9:15 am slot but entered the VPO at 9:20 am. When they entered the office, exactly six out of the ten counters were either processing applications, or had finished processing one and ready to start processing the next.

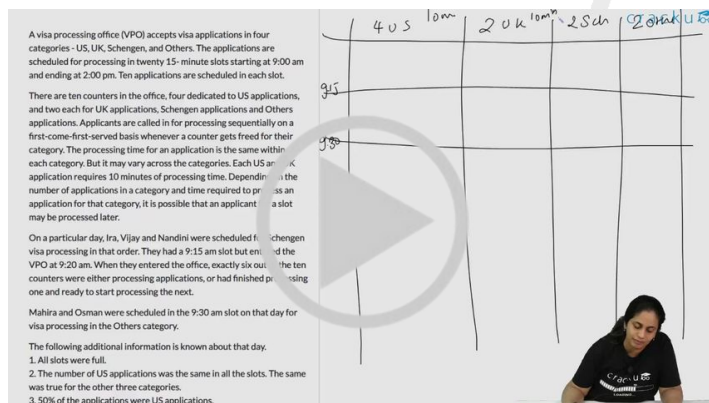
Mahira and Osman were scheduled in the 9:30 am slot on that day for visa processing in the Others category.

The following additional information is known about that day.

1. All slots were full.
2. The number of US applications was the same in all the slots. The same was true for the other three categories.
3. 50% of the applications were US applications.
4. All applicants except Ira, Vijay and Nandini arrived on time.
5. Vijay was called to a counter at 9:25 am.

Question 1

What is the maximum possible value of the total time (in minutes, nearest to its integer value) required to process all applications in the Others category on that day?



A visa processing office (VPO) accepts visa applications in four categories - US, UK, Schengen, and Others. The applications are scheduled for processing in twenty 15- minute slots starting at 9:00 am and ending at 2:00 pm. Ten applications are scheduled in each slot.

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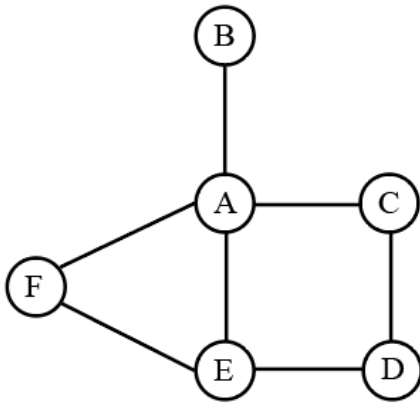
1. All slots were full.
2. The number of US applications was the same in all the slots. The same was true for the other three categories.
3. 50% of the applications were US applications.

VIDEO SOLUTION

Instructions

Comprehension:

A, B, C, D, E and F are the six police stations in an area, which are connected by streets as shown below. Four teams - Team 1, Team 2, Team 3 and Team 4 - patrol these streets continuously between 09:00 hrs. and 12:00 hrs. each day.



The teams need 30 minutes to cross a street connecting one police station to another. All four teams start from Station A at 09:00 hrs. and must return to Station A by 12:00 hrs. They can also pass via Station A at any point on their journeys.

The following facts are known.

1. None of the streets has more than one team traveling along it in any direction at any point in time.
2. Teams 2 and 3 are the only ones in stations E and D respectively at 10:00 hrs.
3. Teams 1 and 3 are the only ones in station E at 10:30 hrs.
4. Teams 1 and 4 are the only ones in stations B and E respectively at 11:30 hrs.
5. Team 1 and Team 4 are the only teams that patrol the street connecting stations A and E.
6. Team 4 never passes through Stations B, D or F.

Question 2

How many times do the teams pass through Station B in a day?

	9	9:30	10	10:30	11	11:30	12
Team 1	A						A
Team 2	A		E				A
Team 3	A		D				A
Team 4	A						A

VIDEO SOLUTION

Instructions

The Humanities department of a college is planning to organize eight seminars, one for each of the eight doctoral students - A, B, C, D, E, F, G and H. Four of them are from Economics, three from Sociology and one from Anthropology department. Each student is guided by one among P, Q, R, S and T. Two students are guided by each of P, R and T, while one student is guided by each of Q and S. Each student is guided by a guide belonging to their department. Each seminar is to be scheduled in one of four consecutive 30-minute slots starting at 9:00 am, 9:30 am, 10:00 am and 10:30 am on the same day. More than one seminars can be scheduled in a slot, provided the guide is free. Only three rooms are available and hence at the most three seminars can be scheduled in a slot. Students who are guided by the same guide must be scheduled in consecutive slots.

The following additional facts are also known.

1. Seminars by students from Economics are scheduled in each of the four slots.
2. A's is the only seminar that is scheduled at 10:00 am. A is guided by R.
3. F is an Anthropology student whose seminar is scheduled at 10:30 am.
4. The seminar of a Sociology student is scheduled at 9:00 am.

5. B and G are both Sociology students, whose seminars are scheduled in the same slot. The seminar of an Economics student, who is guided by T, is also scheduled in the same slot.
6. P, who is guiding both B and C, has students scheduled in the first two slots.
7. A and G are scheduled in two consecutive slots.

Question 3

If D is scheduled in the slot immediately before Q's, then which of the following is NOT necessarily true?

- A G is guided by Q.
- B E is guided by R.
- C F is guided by S.
- D D is guided by T.



Handwritten notes on the left side of the video frame:

- (2) P -
- (1) Q -
- (2) R -
- (1) S -
- (2) T -

A table with four rows and two columns:

9:00	
9:30	
10:00	
10:30	

Text on the right side of the video frame:

10:30 am on the same day. More than one seminars can be scheduled in a slot, provided the guide is free. Only three rooms are available and hence at the most three seminars can be scheduled in a slot. Students who are guided by the same guide must be scheduled in consecutive slots.

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7. A and G are scheduled in two consecutive slots.

A man is sitting at a desk, writing on a notepad.

VIDEO SOLUTION



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Question 4

Who all are NOT guiding any Economics students?

- A Q, R and S
- B P, Q and R
- C P, R and S
- D P, Q and S

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(2) P -
 (1) Q -
 (2) R -
 (1) S -
 (2) T -

9:00	
9:30	
10:00	
10:30	

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 6. The seminar of a student who is guided by both R and C, has students scheduled in the first two slots.
 7. B and G are scheduled in two consecutive slots.

VIDEO SOLUTION

Question 5

Which one of the following statements is true?

- A Two seminars are scheduled in the first slot.
- B Only one seminar is scheduled in the second slot.
- C Three seminars are scheduled in the first slot.
- D Three seminars are scheduled in the last slot.

Cracku

(2) P -
 (1) Q -
 (2) R -
 (1) S -
 (2) T -

9:00	
9:30	
10:00	
10:30	

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Mahira and Osman were scheduled in the 9:30 am slot on that day for visa processing in the Others category.

The following additional information is known about that day.

1. All slots were full.
2. The number of US applications was the same in all the slots. The same was true for the other three categories.
3. 50% of the applications were US applications.
4. All applicants except Ira, Vijay and Nandini arrived on time.
5. Vijay was called to a counter at 9:25 am.

Question 6

When did the application processing for all US applicants get over on that day?

- A 2:05 pm
- B 2:25 pm
- C 2:00 pm
- D 3:40 pm

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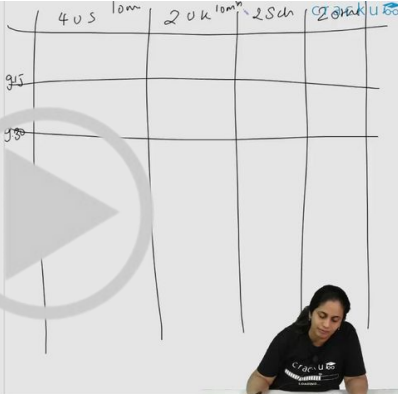
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VIDEO SOLUTION

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Question 7

How many UK applications were scheduled on that day?

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VIDEO SOLUTION

Question 8

Which of the following statements is false?

- A The application process of Osman was completed before 9:45 am.
- B The application process of Mahira started after Nandini's.
- C The application process of Osman was completed before Vijay's.
- D The application process of Mahira was completed before Nandini's.

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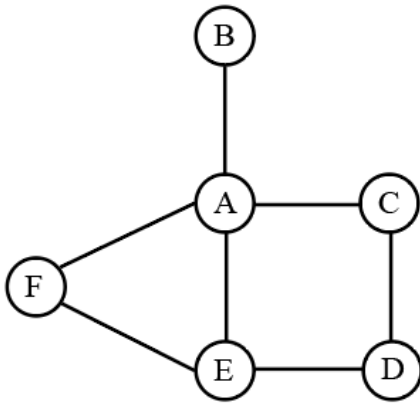
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A, B, C, D, E and F are the six police stations in an area, which are connected by streets as shown below. Four teams - Team 1, Team 2, Team 3 and Team 4 - patrol these streets continuously between 09:00 hrs. and 12:00 hrs. each day.



The teams need 30 minutes to cross a street connecting one police station to another. All four teams start from Station A at 09:00 hrs. and must return to Station A by 12:00 hrs. They can also pass via Station A at any point on their journeys.

The following facts are known.

1. None of the streets has more than one team traveling along it in any direction at any point in time.
2. Teams 2 and 3 are the only ones in stations E and D respectively at 10:00 hrs.
3. Teams 1 and 3 are the only ones in station E at 10:30 hrs.
4. Teams 1 and 4 are the only ones in stations B and E respectively at 11:30 hrs.
5. Team 1 and Team 4 are the only teams that patrol the street connecting stations A and E.
6. Team 4 never passes through Stations B, D or F.

Question 9

How many times does Team 4 pass through Station E in a day?

	9	9:30	10	10:30	11	11:30	12
Team 1	A						A
Team 2	A		E				A
Team 3	A		D				A
Team 4	A						A

VIDEO SOLUTION

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Question 10

How many teams pass through Station C in a day?

- A 4
- B 3
- C 1
- D 2

VIDEO SOLUTION

Instructions

Healthy Bites is a fastfood joint serving three items, burgers, fries and ice cream. It has two employees Anish and Bani who prepare the items ordered by the clients. Preparation time is 10 minutes for a burger and 2 minutes for an order of Ice cream. An employee can prepare only one of these items at a time. The fries are prepared in an automatic fryer which can prepare upto to 3 portions of fries at a time, and takes 5 minutes irrespective of the number of portions. The fryer does not need an employee to constantly attend to it, and we can ignore the time taken by an employee to start and stop the fryer; thus, an employee can be engaged in preparing other items while the frying is on. However fries cannot be prepared in anticipation of future orders.

Healthy Bites wishes to serve the orders as early as possible. The individual items in any order are served as and when ready; however, the order is considered to be completely served only when all the items of that order are served.

The table below gives the orders of three clients and the times at which they placed their orders:

Client no.	Time	Order
1	10:00	1 burger, 3 portions of fries, 1 order of ice cream
2	10:05	2 portions of fries. 1 order of ice cream
3	10:07	1 burger. 1 portion of fries

Question 11

Assume that only one client's order can be processed at any given point of time. So, Anish or Bani cannot start preparing a new order while a previous order is being prepared.

At what time is the order placed by Client 1 completely served?

- A 10:17
- B 10:10
- C 10:15
- D 10:20

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Assume that only one client's order can be processed at any given point of time. So, Anish or Bani cannot start preparing a new order while a previous order is being prepared.
At what time is the order placed by Client 1 completely served?

+




VIDEO SOLUTION

Question 12

Suppose the employees are allowed to process multiple orders at a time, but the preference would be to finish orders of clients who placed their orders earlier.

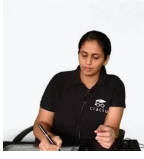
Also assume that the fourth client came in only at 10:35. Between 10:00 and 10:30, for how many minutes is exactly one of the employees idle?

- A 7
- B 10
- C 15
- D 23

cracku

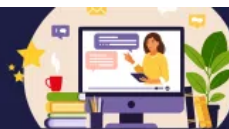
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+

VIDEO SOLUTION

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Instructions

A journal plans to publish 18 research papers, written by eight authors (A, B, C, D, E, F, G, and H) in four issues of the journal scheduled in January, April, July and October. Each of the research papers was written by exactly one of the eight authors. Five papers were scheduled in each of the first two issues, while four were scheduled in each of the last two issues. Every author wrote at least one paper and at most three papers. The total number of papers written by A, D, G and H was double the total number of papers written by the other four authors. Four

of the authors were from India and two each were from Japan and China. Each author belonged to exactly one of the three areas – Manufacturing, Automation, and Logistics. Four of the authors were from the Logistics area and two were from the Automation area. As per the journal policy, none of the authors could have more than one paper in any issue of the journal.

The following facts are also known.

1. F, an Indian author from the Logistics area, wrote only one paper. It was scheduled in the October issue.
2. A was from the Automation area and did not have a paper scheduled in the October issue.
3. None of the Indian authors were from the Manufacturing area and none of the Japanese or Chinese authors were from the Automation area.
4. A and H were from different countries, but had their papers scheduled in exactly the same issues.
5. C and E, both Chinese authors from different areas, had the same number of papers scheduled. Further, E had papers scheduled in consecutive issues of the journal but C did not.
6. B, from the Logistics area, had a paper scheduled in the April issue of the journal.
7. B and G belonged to the same country. None of their papers were scheduled in the same issue of the journal.
8. D, a Japanese author from the Manufacturing area, did not have a paper scheduled in the July issue.
9. C and H belonged to different areas.

Question 13

What is the correct sequence of number of papers written by B, C, E and G, respectively?

- A 1, 2, 2, 3
- B 1, 3, 3, 1
- C 3, 1, 1, 3
- D 1, 2, 2, 1

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$A+D+G+H = 2(B+C+E+F)$

$2x + 11 = 2(2x + 18)$

$2x + 11 = 4x + 36$

$-2x = 25$

$x = -12.5$

Jan - 5
April - 5
July - 4
Oct - 4

$x + 2x = 218$

$2x = 218$

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8. D, a Japanese author from the Manufacturing area, did not have a paper scheduled in the July issue.
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VIDEO SOLUTION

Question 14

Which of the following statement(s) MUST be true?

- Statement A: Every issue had at least one paper by author(s) from each country.
- Statement B: Every issue had at most two papers by author(s) from each area.

- A Both the statements
- B Only Statement B
- C Only Statement A
- D Neither of the statements

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$$A + D + G + H = 2(B + E + F)$$

Jan - 5
April - 5
July - 4
Oct - 4
Total = 18

$x + 2x = 18$
 $3x = 18$
 $x = 6$

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8. F, from the Logistics area, had a paper scheduled in the January issue of the journal.

VIDEO SOLUTION

Question 15

Which of the following statements is FALSE?

- A Every issue had at least one paper by author(s) from Automation area.
- B Every issue had exactly one paper by a Chinese author.
- C Every issue had exactly two papers by authors from Logistics area.
- D Every issue had exactly two papers by Indian authors.

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$$A + D + G + H = 2(B + E + F)$$

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2. One author from the Automation area and did not have a paper scheduled in the October issue.

3. None of the Indian authors were from the Manufacturing area and none of the Japanese or Chinese authors were from the Automation area.

4. A and H were from different countries, but had their papers scheduled in exactly the same issue.

5. C and E, both Chinese authors from different areas, had the same number of papers scheduled. Further, E had papers scheduled in consecutive issues of the journal but C did not.

6. B, from the Logistics area, had a paper scheduled in the April issue of the journal.

7. B and G belonged to the same country. None of their papers were scheduled in the same issue of the journal.

8. F, from the Logistics area, had a paper scheduled in the January issue of the journal.

VIDEO SOLUTION

CAT 2024 Coaching By 4 Time CAT 100%iler & IIM Alumni

Question 16

Which of the following is the correct sequence of number of papers by authors from Automation, Manufacturing and Logistics areas, respectively?

- A 6, 5, 7
- B 6, 6, 6
- C 6, 7, 5
- D 5, 6, 7

cracku

$A + D + G + H = 2(B + C + E + F)$

$2x = 12$

$x = 6$

Jan - 5

April - 5

July - 4

Oct - 4

$x + 2x = 18$

$2x = 18$

A journal plans to publish 18 research papers, written by eight authors (A, B, C, D, E, F, G, and H) in four issues of the journal scheduled in January, April, July and October. Each of the research papers was written by exactly one of the eight authors. Five papers were scheduled in each of the first two issues, while four were scheduled in each of the last two issues. Every author wrote at least one paper and at most three papers. The total number of papers written by A, D, G and H was double the total number of papers written by the other four authors. Four of the authors were from India and two were from Japan and China. Each author belonged to exactly one of the three areas: Manufacturing, Automation, and Logistics. Four of the authors were from the Logistics area and two were from the Automation area. As per the journal policy, none of the authors could have more than one paper in any issue of the journal.

1. The authors are also known. The author from the Logistics area, wrote only one paper. It was scheduled in the first issue. The author from the Automation area and did not have a paper scheduled in the first issue.

2. None of the Indian authors were from the Manufacturing area and none of the Japanese or Chinese authors were from the Automation area.

3. A and H were from different countries, but had their papers scheduled in exactly the same issue.

4. C and E, both Chinese authors from different areas, had the same number of papers scheduled. Further, E had papers scheduled in consecutive issues of the journal but C did not.

5. B, from the Logistics area, had a paper scheduled in the April issue of the journal.

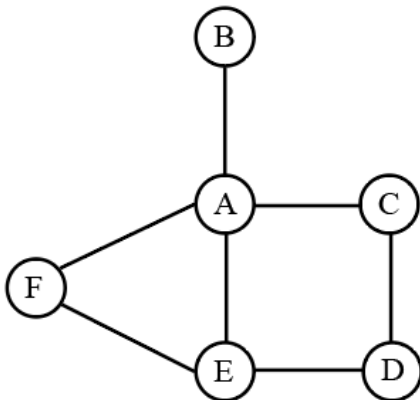
6. B and G belonged to the same country. None of their papers were scheduled in the same issue of the journal.

VIDEO SOLUTION

Instructions

Comprehension:

A, B, C, D, E and F are the six police stations in an area, which are connected by streets as shown below. Four teams - Team 1, Team 2, Team 3 and Team 4 - patrol these streets continuously between 09:00 hrs. and 12:00 hrs. each day.



The teams need 30 minutes to cross a street connecting one police station to another. All four teams start from Station A at 09:00 hrs. and must return to Station A by 12:00 hrs. They can also pass via Station A at any point on their journeys.

The following facts are known.

1. None of the streets has more than one team traveling along it in any direction at any point in time.
2. Teams 2 and 3 are the only ones in stations E and D respectively at 10:00 hrs.
3. Teams 1 and 3 are the only ones in station E at 10:30 hrs.
4. Teams 1 and 4 are the only ones in stations B and E respectively at 11:30 hrs.
5. Team 1 and Team 4 are the only teams that patrol the street connecting stations A and E.
6. Team 4 never passes through Stations B, D or F.

Question 17

Which team patrols the street connecting Stations D and E at 10:15 hrs?

- A Team 4
- B Team 1
- C Team 2
- D Team 3

	9	9:50	10	10:30	11	11:30	12
Team 1	A		E				A
Team 2	A		E				A
Team 3	A		D				A
Team 4	A						A

The teams need to cross a street connecting one police station to another. All four teams start at Station A at 09:00 hrs. and must return to Station A by 12:00 hrs. They can only move along the streets shown in the diagram. The diagram shows a network of stations (A, B, C, D, E, F) connected by streets. The text describes the movement of four teams (1, 2, 3, 4) starting from Station A at 09:00 hrs. and returning to Station A by 12:00 hrs. It details their paths through various streets and stations.

VIDEO SOLUTION

Instructions

A visa processing office (VPO) accepts visa applications in four categories - US, UK, Schengen, and Others. The applications are scheduled for processing in twenty 15- minute slots starting at 9:00 am and ending at 2:00 pm. Ten applications are scheduled in each slot.

There are ten counters in the office, four dedicated to US applications, and two each for UK applications, Schengen applications and Others applications. Applicants are called in for processing sequentially on a first-come-first-served basis whenever a counter gets freed for their category. The processing time for an application is the same within each category. But it may vary across the categories. Each US and UK application requires 10 minutes of processing time. Depending on the number of applications in a category and time required to process an application for that category, it is possible that an applicant for a slot may be processed later.

On a particular day, Ira, Vijay and Nandini were scheduled for Schengen visa processing in that order. They had a 9:15 am slot but entered the VPO at 9:20 am. When they entered the office, exactly six out of the ten counters were either processing applications, or had finished processing one and ready to start processing the next.

Mahira and Osman were scheduled in the 9:30 am slot on that day for visa processing in the Others category.

The following additional information is known about that day.

1. All slots were full.
2. The number of US applications was the same in all the slots. The same was true for the other three categories.
3. 50% of the applications were US applications.
4. All applicants except Ira, Vijay and Nandini arrived on time.
5. Vijay was called to a counter at 9:25 am.

Question 18

Which of the following is the closest to the time when Nandini's application process got over?

- A 9:50 am
- B 9:37 am
- C 9:35 am
- D 9:45 am

A visa processing office (VPO) accepts visa applications in four categories - US, UK, Schengen, and Others. The applications are scheduled for processing in twenty 15-minute slots starting at 9:00 am and ending at 2:00 pm. Ten applications are scheduled in each slot.

There are ten counters in the office, four dedicated to US applications, and two each for UK applications, Schengen applications and Others applications. Applicants are called in for processing sequentially on a first-come-first-served basis whenever a counter gets freed for their category. The processing time for an application is the same within each category. But it may vary across the categories. Each US application requires 10 minutes of processing time. Depending on the number of applications in a category and time required to process an application for that category, it is possible that an applicant's slot may be processed later.

On a particular day, Ira, Vijay and Nandini were scheduled for Schengen visa processing in that order. They had a 9:15 am slot but entered the VPO at 9:20 am. When they entered the office, exactly six out of the ten counters were either processing applications, or had finished processing one and ready to start processing the next.

Mahira and Osman were scheduled in the 9:30 am slot on that day for visa processing in the Others category.

The following additional information is known about that day.

1. All slots were full.
2. The number of US applications was the same in all the slots. The same was true for the other three categories.
3. 50% of the applications were US applications.

VIDEO SOLUTION

CAT Previous Papers PDF

Instructions

Healthy Bites is a fastfood joint serving three items, burgers, fries and ice cream. It has two employees Anish and Bani who prepare the items ordered by the clients. Preparation time is 10 minutes for a burger and 2 minutes for an order of Ice cream. An employee can prepare only one of these items at a time. The fries are prepared in an automatic fryer which can prepare upto to 3 portions of fries at a time, and takes 5 minutes irrespective of the number of portions. The fryer does not need an employee to constantly attend to it, and we can ignore the time taken by an employee to start and stop the fryer; thus, an employee can be engaged in preparing other items while the frying is on. However fries cannot be prepared in anticipation of future orders.

Healthy Bites wishes to serve the orders as early as possible. The individual items in any order are served as and when ready; however, the order is considered to be completely served only when all the items of that order are served.

The table below gives the orders of three clients and the times at which they placed their orders:

Client no.	Time	Order
1	10:00	1 burger, 3 portions of fries, 1 order of ice cream
2	10:05	2 portions of fries. 1 order of ice cream
3	10:07	1 burger. 1 portion of fries

Question 19

Assume that only one client's order can be processed at any given point of time. So, Anish or Bani cannot start preparing a new order while a previous order is being prepared.

At what time is the order placed by Client 3 completely served?

- A 10:35
- B 10:22
- C 10:25
- D 10:17

cracku

Assume that only one client's order can be processed at any given point of time. So, Anish or Bani cannot start preparing a new order while a previous order is being prepared.
At what time is the order placed by Client 1 completely served?

+



VIDEO SOLUTION

Question 20

Suppose the employees are allowed to process multiple orders at a time, but the preference would be to finish orders of clients who placed their orders earlier.

At what time is the order placed by Client 2 completely served?

- A 10:10
- B 10:12
- C 10:15
- D 10:17

cracku

Assume that only one client's order can be processed at any given point of time. So, Anish or Bani cannot start preparing a new order while a previous order is being prepared.
At what time is the order placed by Client 1 completely served?

+



VIDEO SOLUTION

Instructions

Three doctors, Dr. Ben, Dr. Kane and Dr. Wayne visit a particular clinic Monday to Saturday to see patients. Dr. Ben sees each patient for 10 minutes and charges Rs. 100/-. Dr. Kane sees each patient for 15 minutes and charges Rs. 200/-, while Dr. Wayne sees each patient for 25 minutes and charges Rs. 300/-. The clinic has three rooms numbered 1, 2 and 3 which are assigned to the three doctors as per the following table.

Room No.	Monday & Tuesday	Wednesday & Thursday	Friday & Saturday
1	Ben	Wayne	Kane
2	Kane	Ben	Wayne
3	Wayne	Kane	Ben

The clinic is open from 9 a.m. to 11.30 a.m. every Monday to Saturday.

On arrival each patient is handed a numbered token indicating their position in the queue, starting with token number 1 every day. As soon as any doctor becomes free, the next patient in the queue enters that emptied room for consultation. If at any time, more than one room is free then the waiting patient enters the room with the smallest number. For example, if the next two patients in the queue have token numbers 7 and 8 and if rooms numbered 1 and 3 are free, then patient with token number 7 enters room number 1 and patient with token number 8 enters room number 3.

Question 21

The queue is never empty on one particular Saturday. Which of the three doctors would earn the maximum amount in consultation charges on that day?

- A Dr. Wayne
- B Dr. Kane
- C Dr. Ben
- D Both Dr. Wayne and Dr. Kane

cracku

while Dr. Wayne sees each patient for 25 minutes and charges Rs. 300/-. The clinic has three rooms numbered 1, 2 and 3 which are assigned to the three doctors as per the following table.

Room No.	Monday & Tuesday	Wednesday & Thursday	Friday & Saturday
1	Ben	Wayne	Kane
2	Kane	Ben	Wayne
3	Wayne	Kane	Ben

The clinic is open from 9 a.m. to 11.30 a.m. every Monday to Saturday.

On arrival each patient is handed a numbered token indicating their position in the queue, starting with token number 1 every day. As soon as any doctor becomes free, the next patient in the queue enters that emptied room for consultation. If at any time, more than one room is free then the waiting patient enters the room with the smallest number. For example, if the next two patients in the queue have token numbers 7 and 8 and if rooms numbered 1 and 3 are free, then patient with token number 7 enters room number 1 and patient with token number 8 enters room number 3.

Question (1)
What is the maximum number of patients that the clinic can cater to on any single day?

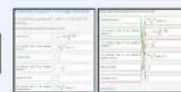
9-11:30
b = 150 - 15 = 135



VIDEO SOLUTION



CAT FORMULAS REVISION



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Question 22

Mr. Singh visited the clinic on Monday, Wednesday, and Friday of a particular week, arriving at 8:50 a.m. on each of the three days. His token number was 13 on all three days. On which day was he at the clinic for the maximum duration?

- A Same duration on all three days
- B Friday
- C Monday
- D Wednesday

Cracku

While Dr. Wayne sees each patient for 25 minutes and charges Rs. 300/-. The clinic has three rooms numbered 1, 2 and 3 which are assigned to the three doctors as per the following table.

Room No.	Monday & Tuesday	Wednesday & Thursday	Friday & Saturday
1	Ben	Wayne	Kane
2	Kane	Ben	Wayne
3	Wayne	Kane	Ben

The clinic is open from 9 a.m. to 11.30 a.m. every Monday to Saturday.

On arrival each patient is handed a numbered token indicating their position in the queue, starting with token number 1 every day. As soon as any doctor becomes free, the next patient in the queue enters that emptied room for consultation. If at any time, more than one room is free then the waiting patient enters the room with the smallest number. For example, if the next two patients in the queue have token numbers 7 and 8 and if rooms numbered 1 and 3 are free, then patient with token number 7 enters room number 1 and patient with token number 8 enters room number 3.

Question (1)

What is the maximum number of patients that the clinic can cater to on any single day?

Handwritten notes: 9-11:30, 25 min, $b = 150 = 15$



VIDEO SOLUTION

Question 23

On a slow Thursday, only two patients are waiting at 9 a.m. After that two patients keep arriving at exact 15-minute intervals starting at 9:15 a.m. – i.e. at 9:15 a.m., 9:30 a.m., 9:45 a.m. etc. Then the total duration in minutes when all three doctors are simultaneously free is

- A 30
- B 10
- C 15
- D 0

Cracku

While Dr. Wayne sees each patient for 25 minutes and charges Rs. 300/-. The clinic has three rooms numbered 1, 2 and 3 which are assigned to the three doctors as per the following table.

Room No.	Monday & Tuesday	Wednesday & Thursday	Friday & Saturday
1	Ben	Wayne	Kane
2	Kane	Ben	Wayne
3	Wayne	Kane	Ben

The clinic is open from 9 a.m. to 11.30 a.m. every Monday to Saturday.

On arrival each patient is handed a numbered token indicating their position in the queue, starting with token number 1 every day. As soon as any doctor becomes free, the next patient in the queue enters that emptied room for consultation. If at any time, more than one room is free then the waiting patient enters the room with the smallest number. For example, if the next two patients in the queue have token numbers 7 and 8 and if rooms numbered 1 and 3 are free, then patient with token number 7 enters room number 1 and patient with token number 8 enters room number 3.

Question (1)

What is the maximum number of patients that the clinic can cater to on any single day?

Handwritten notes: 9-11:30, 25 min, $b = 150 = 15$



VIDEO SOLUTION

Instructions

The Humanities department of a college is planning to organize eight seminars, one for each of the eight doctoral students - A, B, C, D, E, F, G and H. Four of them are from Economics, three from Sociology and one from Anthropology department. Each student is guided by one among P, Q, R, S and T. Two students are guided by each of P, R and T, while one student is guided by each of Q and S. Each student is guided by a guide belonging to their department. Each seminar is to be scheduled in one of four consecutive 30-minute slots starting at 9:00 am, 9:30 am, 10:00 am and 10:30 am on the same day. More than one seminars can be scheduled in a slot, provided the guide is free. Only three rooms are available and hence at the most three seminars can be scheduled in a slot. Students who are guided by the same guide must be scheduled in consecutive slots.

The following additional facts are also known.

1. Seminars by students from Economics are scheduled in each of the four slots.
2. A's is the only seminar that is scheduled at 10:00 am. A is guided by R.
3. F is an Anthropology student whose seminar is scheduled at 10:30 am.
4. The seminar of a Sociology student is scheduled at 9:00 am.
5. B and G are both Sociology students, whose seminars are scheduled in the same slot. The seminar of an Economics student, who is guided by T, is also scheduled in the same slot.
6. P, who is guiding both B and C, has students scheduled in the first two slots.
7. A and G are scheduled in two consecutive slots.

Question 24

Which of the following statements is necessarily true?

- A Q is guiding G.
- B H is an Economics student.
- C S is guiding F.
- D B is scheduled in the first slot.



B -
 C -
 D -
 E -
 F -
 G -
 H -

(2) P -
 (1) Q -
 (2) R -
 (1) S -
 (2) T -

9:00	
9:30	
10:00	
10:30	

10:30 am on the same day. More than one seminars can be scheduled in a slot, provided the guide is free. Only three rooms are available and hence at the most three seminars can be scheduled in a slot. Students who are guided by the same guide must be scheduled in consecutive slots.
 The following additional facts are also known.
 1. Seminars by students from Economics are scheduled in each of the four slots.
 2. A's is the only seminar that is scheduled at 10:00 am. A is guided by R.
 3. F is an Anthropology student whose seminar is scheduled at 10:30 am.
 4. The seminar of a Sociology student is scheduled at 9:00 am.
 5. B and G are both Sociology students, whose seminars are scheduled in the same slot. The seminar of an Economics student, who is guided by T, is also scheduled in the same slot.
 6. P, who is guiding both B and C, has students scheduled in the first two slots.
 7. A and G are scheduled in two consecutive slots.



VIDEO SOLUTION

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Question 25

If D is scheduled in a slot later than Q's, then which of the following two statement(s) is(are) true?

- (i) E and H are guided by T.
- (ii) G is guided by Q.

- A Only (ii)
- B Neither (i) nor (ii)
- C Only (i)
- D Both (i) and (ii)

Cracku

(2) P -
(1) Q -
(2) R -
(1) S -
(2) T -

9:00	
9:30	
10:00	
10:30	

10:30 am on the same day. More than one seminars can be scheduled in a slot, provided the guide is free. Only three rooms are available and hence at the most three seminars can be scheduled in a slot. Students who are guided by the same guide must be scheduled in consecutive slots.

The following additional facts are also known.

1. Seminars by students from Economics are scheduled in each of the four slots.
2. A's is the only seminar that is scheduled at 10:00 am. A is guided by R.
3. F is an Anthropology student whose seminar is scheduled at 10:30 am.
4. The seminar of a Biology student is scheduled at 9:00 am.
5. B and G are both Sociology students, whose seminars are scheduled in the same slot. The seminar of an Economics student, who is guided by T, is scheduled in the same slot.
6. The seminar of a student who is guided by both R and C, has students scheduled in the first two slots.
7. B and G are scheduled in two consecutive slots.

VIDEO SOLUTION

Question 26

If E and Q are both scheduled in the same slot, then which of the following statements BEST describes the relationship between D, H, and T?

- A Exactly one of D and H is guided by T.
- B Both D and H are guided by T.
- C At least one of D and H is guided by T.
- D Neither D nor H is guided by T.

Cracku

(2) P -
(1) Q -
(2) R -
(1) S -
(2) T -

9:00	
9:30	
10:00	
10:30	

10:30 am on the same day. More than one seminars can be scheduled in a slot, provided the guide is free. Only three rooms are available and hence at the most three seminars can be scheduled in a slot. Students who are guided by the same guide must be scheduled in consecutive slots.

The following additional facts are also known.

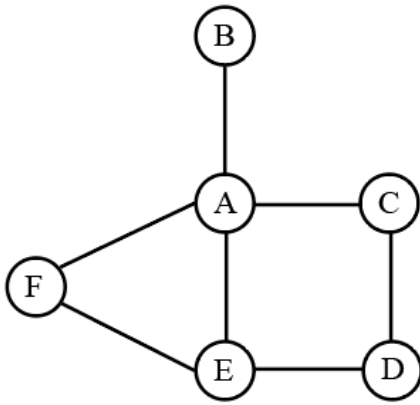
1. Seminars by students from Economics are scheduled in each of the four slots.
2. A's is the only seminar that is scheduled at 10:00 am. A is guided by R.
3. F is an Anthropology student whose seminar is scheduled at 10:30 am.
4. The seminar of a Biology student is scheduled at 9:00 am.
5. B and G are both Sociology students, whose seminars are scheduled in the same slot. The seminar of an Economics student, who is guided by T, is scheduled in the same slot.
6. The seminar of a student who is guided by both R and C, has students scheduled in the first two slots.
7. B and G are scheduled in two consecutive slots.

VIDEO SOLUTION

Instructions

Comprehension:

A, B, C, D, E and F are the six police stations in an area, which are connected by streets as shown below. Four teams - Team 1, Team 2, Team 3 and Team 4 - patrol these streets continuously between 09:00 hrs. and 12:00 hrs. each day.



The teams need 30 minutes to cross a street connecting one police station to another. All four teams start from Station A at 09:00 hrs. and must return to Station A by 12:00 hrs. They can also pass via Station A at any point on their journeys.

The following facts are known.

1. None of the streets has more than one team traveling along it in any direction at any point in time.
2. Teams 2 and 3 are the only ones in stations E and D respectively at 10:00 hrs.
3. Teams 1 and 3 are the only ones in station E at 10:30 hrs.
4. Teams 1 and 4 are the only ones in stations B and E respectively at 11:30 hrs.
5. Team 1 and Team 4 are the only teams that patrol the street connecting stations A and E.
6. Team 4 never passes through Stations B, D or F.

Question 27

Which one among the following stations is visited the largest number of times?

- A Station C
- B Station E
- C Station D
- D Station F

	9	9:30	10	10:30	11	11:30	12
Team 1	A						A
Team 2	A		E				A
Team 3	A		D				A
Team 4	A						A

The teams need 30 minutes to cross a street connecting one police station to another. All four teams start from Station A at 09:00 hrs. and must return to Station A by 12:00 hrs. They can also pass via Station A at any point on their journeys.

The following facts are known:

1. None of the streets has more than one team traveling along it in any direction at any point in time.
2. Teams 2 and 3 are the only ones in stations E and D respectively at 10:00 hrs.
3. Teams 1 and 3 are the only ones in station E at 10:30 hrs.
4. Teams 1 and 4 are the only ones in stations B and E respectively at 11:30 hrs.
5. Team 1 and Team 4 are the only teams that patrol the street connecting stations A and E.
6. Team 4 never passes through Stations B, D or F.

VIDEO SOLUTION

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CAT 2024 SYLLABUS



Instructions

Seven candidates, Akil, Balaram, Chitra, Divya, Erina, Fatima, and Ganeshan, were invited to interview for a position. Candidates were required to reach the venue before 8 am. Immediately upon arrival, they were sent to one of three interview rooms: 101, 102, and 103. The following venue log shows the arrival times for these candidates. Some of the names have not been recorded in the log and have been marked as '?'.

Time	7:10 AM	7:15 AM	7:25 AM	7:30 AM	7:40 AM	7:45 AM
Person	Akil, ?	?	?	Chitra	Fatima	?

Additionally here are some statements from the candidates:

Balaram: I was the third person to enter Room 101.

Chitra: I was the last person to enter the room I was allotted to.

Erina: I was the only person in the room I was allotted to.

Fatima: Three people including Akhil were already in the room that I was allotted to when I entered it.

Ganeshan : I was one among the two candidates allotted to Room 102.

Question 28

When did Erina reach the venue?

- A 7:45 am
- B 7:25 am
- C 7:15 am
- D 7:10 am

VIDEO SOLUTION

Question 29

Who else was in Room 102 when Ganeshan entered?

- A Akil
- B Divya
- C Chitra
- D No one

cracku

Seven candidates, Akh, Balam, Chitra, Divya, Erina, Fatima, and Ganesha, were invited to interview for a position. Candidates were required to reach the venue before 8 am. Immediately upon arrival, they were sent to one of three interview rooms: 101, 102, and 103. The following venue log shows the arrival times for these candidates. Some of the names have not been recorded in the log and have been marked as "?".

Time	17:10 AM	7:15 AM	7:20 AM	7:30 AM	7:40 AM	7:45 AM
Person	Akh	?	?	?	Chitra	?

Additionally here are some statements from the candidates:

- Balam: I was the third person to enter Room 101.
- Chitra: I was the last person to enter the room I was allotted to.
- Erina: I was the only person in the room I was allotted to.
- Fatima: Three people including Akh were already in the room that I was allotted to when I entered it.
- Ganesha: I was one among the two candidates allotted to Room 102.

Time

A - 7:10

B -

C - 7:30

D -

E -

G -

101

102

103

x101

40

VIDEO SOLUTION

Question 30

What best can be said about the room to which Divya was allotted?

- A Definitely Room 101
- B Definitely Room 103
- C Definitely Room 102
- D Either Room 101 or Room 102

cracku

Seven candidates, Akh, Balam, Chitra, Divya, Erina, Fatima, and Ganesha, were invited to interview for a position. Candidates were required to reach the venue before 8 am. Immediately upon arrival, they were sent to one of three interview rooms: 101, 102, and 103. The following venue log shows the arrival times for these candidates. Some of the names have not been recorded in the log and have been marked as "?".

Time	17:10 AM	7:15 AM	7:20 AM	7:30 AM	7:40 AM	7:45 AM
Person	Akh	?	?	?	Chitra	?

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- Erina: I was the only person in the room I was allotted to.
- Fatima: Three people including Akh were already in the room that I was allotted to when I entered it.
- Ganesha: I was one among the two candidates allotted to Room 102.

Time

A - 7:10

B -

C - 7:30

D -

E -

G -

101

102

103

x101

40

VIDEO SOLUTION

[CAT Formulas PDF \[Download Now\]](#)

Instructions

A journal plans to publish 18 research papers, written by eight authors (A, B, C, D, E, F, G, and H) in four issues of the journal scheduled in January, April, July and October. Each of the research papers was written by exactly one of the eight authors. Five papers were scheduled in each of the first two issues, while four were scheduled in each of the last two issues. Every author wrote at least one paper and at most three papers. The total number of papers written by A, D, G and H was double the total number of papers written by the other four authors. Four of the authors were from India and two each were from Japan and China. Each author belonged to exactly one of the three areas — Manufacturing, Automation, and Logistics. Four of the authors were from the Logistics area and two were from the Automation area. As per the journal policy, none of the authors could have more than one paper in any issue of the journal.

The following facts are also known.

1. F, an Indian author from the Logistics area, wrote only one paper. It was scheduled in the October issue.
2. A was from the Automation area and did not have a paper scheduled in the October issue.
3. None of the Indian authors were from the Manufacturing area and none of the Japanese or Chinese authors were from the Automation area.
4. A and H were from different countries, but had their papers scheduled in exactly the same issues.
5. C and E, both Chinese authors from different areas, had the same number of papers scheduled. Further, E had papers scheduled in consecutive issues of the journal but C did not.
6. B, from the Logistics area, had a paper scheduled in the April issue of the journal.
7. B and G belonged to the same country. None of their papers were scheduled in the same issue of the journal.
8. D, a Japanese author from the Manufacturing area, did not have a paper scheduled in the July issue.
9. C and H belonged to different areas.

Question 31

Which of the following statements is FALSE?

- A There were exactly two papers by authors from Manufacturing area in the January issue.
- B There was exactly one paper by an author from Manufacturing area in the April issue.
- C There was exactly one paper by an author from Logistics area in the October issue.
- D There were exactly two papers by authors from Manufacturing area in the July issue.

cracku

$A+D+G+H = 2(B+E+F)$

$2x + 11 = 2(6 + 4 + 4)$

$2x + 11 = 26$

$2x = 15$

$x = 7.5$

Jan - 5
April - 5
July - 4
Oct - 4

$x + 2x = 18$

$3x = 18$

$x = 6$

A journal plans to publish 18 research papers, written by eight authors (A, B, C, D, E, F, G, and H) in four issues of the journal scheduled in January, April, July and October. Each of the research papers was written by exactly one of the eight authors. Five papers were scheduled in each of the first two issues, while four were scheduled in each of the last two issues. Every author wrote at least one paper and at most three papers. The total number of papers written by A, D, G and H was double the total number of papers written by the other four authors. Four of the authors were from India and two were from Japan and China. Each author belonged to exactly one of the three areas: Manufacturing, Automation, and Logistics. Four of the authors were from the Logistics area and two were from the Automation area. As per the journal policy, none of the authors could have more than one paper in any issue of the journal.

The following facts are also known:

1. F, an Indian author from the Logistics area, wrote only one paper. It was scheduled in the October issue.
2. A was from the Automation area and did not have a paper scheduled in the October issue.
3. None of the Indian authors were from the Manufacturing area and none of the Japanese or Chinese authors were from the Automation area.
4. A and H were from different countries, but had their papers scheduled in exactly the same issues.
5. C and E, both Chinese authors from different areas, had the same number of papers scheduled. Further, E had papers scheduled in consecutive issues of the journal but C did not.
6. B, from the Logistics area, had a paper scheduled in the April issue of the journal.
7. B and G belonged to the same country. None of their papers were scheduled in the same issue of the journal.
8. D, a Japanese author from the Manufacturing area, did not have a paper scheduled in the July issue.
9. C and H belonged to different areas.

VIDEO SOLUTION

Question 32

How many papers were written by Indian authors?

cracku

$A+D+G+H = 2(B+E+F)$

$2x + 11 = 2(6 + 4 + 4)$

$2x + 11 = 26$

$2x = 15$

$x = 7.5$

Jan - 5
April - 5
July - 4
Oct - 4

$x + 2x = 18$

$3x = 18$

$x = 6$

A journal plans to publish 18 research papers, written by eight authors (A, B, C, D, E, F, G, and H) in four issues of the journal scheduled in January, April, July and October. Each of the research papers was written by exactly one of the eight authors. Five papers were scheduled in each of the first two issues, while four were scheduled in each of the last two issues. Every author wrote at least one paper and at most three papers. The total number of papers written by A, D, G and H was double the total number of papers written by the other four authors. Four of the authors were from India and two were from Japan and China. Each author belonged to exactly one of the three areas: Manufacturing, Automation, and Logistics. Four of the authors were from the Logistics area and two were from the Automation area. As per the journal policy, none of the authors could have more than one paper in any issue of the journal.

The following facts are also known:

1. F, an Indian author from the Logistics area, wrote only one paper. It was scheduled in the October issue.
2. A was from the Automation area and did not have a paper scheduled in the October issue.
3. None of the Indian authors were from the Manufacturing area and none of the Japanese or Chinese authors were from the Automation area.
4. A and H were from different countries, but had their papers scheduled in exactly the same issues.
5. C and E, both Chinese authors from different areas, had the same number of papers scheduled. Further, E had papers scheduled in consecutive issues of the journal but C did not.
6. B, from the Logistics area, had a paper scheduled in the April issue of the journal.
7. B and G belonged to the same country. None of their papers were scheduled in the same issue of the journal.
8. D, a Japanese author from the Manufacturing area, did not have a paper scheduled in the July issue.
9. C and H belonged to different areas.

VIDEO SOLUTION

Instructions

Three doctors, Dr. Ben, Dr. Kane and Dr. Wayne visit a particular clinic Monday to Saturday to see patients. Dr. Ben sees each patient for 10 minutes and charges Rs. 100/-. Dr. Kane sees each patient for 15 minutes and charges Rs. 200/-, while Dr. Wayne sees each patient for 25 minutes and charges Rs. 300/-. The clinic has three rooms numbered 1, 2 and 3 which are assigned to the three doctors as per the following table.

Room No.	Monday & Tuesday	Wednesday & Thursday	Friday & Saturday
1	Ben	Wayne	Kane
2	Kane	Ben	Wayne
3	Wayne	Kane	Ben

The clinic is open from 9 a.m. to 11.30 a.m. every Monday to Saturday.

On arrival each patient is handed a numbered token indicating their position in the queue, starting with token number 1 every day. As soon as any doctor becomes free, the next patient in the queue enters that emptied room for consultation. If at any time, more than one room is free then the waiting patient enters the room with the smallest number. For example, if the next two patients in the queue have token numbers 7 and 8 and if rooms numbered 1 and 3 are free, then patient with token number 7 enters room number 1 and patient with token number 8 enters room number 3.

Question 33

What is the maximum number of patients that the clinic can cater to on any single day?

- A 12
- B 30
- C 31
- D 15

Question (1)
What is the maximum number of patients that the clinic can cater to on any single day?

9 - 11:30
2.5 hours
2.5 hours = 150 minutes

VIDEO SOLUTION



CAT FORMULAS BOOK
BY 100%ILER



Instructions

Seven candidates, Akil, Balaram, Chitra, Divya, Erina, Fatima, and Ganeshan, were invited to interview for a position. Candidates were required to reach the venue before 8 am. Immediately upon arrival, they were sent to one of three interview rooms: 101, 102, and 103. The following venue log shows the arrival times for these candidates. Some of the names have not been recorded in the log and have been marked as '?'.
101
102
103

Time	7:10 AM	7:15 AM	7:25 AM	7:30 AM	7:40 AM	7:45 AM
Person	Akil, ?	?	?	Chitra	Fatima	?

Additionally here are some statements from the candidates:

Balaram: I was the third person to enter Room 101.

Chitra: I was the last person to enter the room I was allotted to.

Erina: I was the only person in the room I was allotted to.

Fatima: Three people including Akhil were already in the room that I was allotted to when I entered it.

Ganeshan : I was one among the two candidates allotted to Room 102.

Question 34

If Ganeshan entered the venue before Divya, when did Balaram enter the venue?

- A 7:25 am
- B 7:10 am
- C 7:15 am
- D 7:45 am

cracku

Seven candidates, Akil, Balaram, Chitra, Divya, Erina, Fatima, and Ganeshan, were invited to interview for a position. Candidates were required to reach the venue before 8 am. Immediately upon arrival, they were sent to one of three interview rooms: 101, 102 and 103. The following venue log shows the arrival times for these candidates. Some of the names have not been recorded in the log and have been marked as "?".

Time	7:10 AM	7:15 AM	7:20 AM	7:30 AM	7:40 AM	7:45 AM
Person	Akil, ?	?	?	Chitra	Fatima	?

Additionally here are some statements from the candidate

Balaram: I was the third person to enter Room 101.

Chitra: I was the last person to enter the room I was allotted to.

Erina: I was the only person in the room I was allotted to.

Fatima: Three people including Akhil were already in the room that I was allotted to when I entered it.

Ganeshan: I was one among the two candidates allotted to Room 102.

Handwritten notes on the video screen:

Time

A - 7:10

B -

C - 7:30

D -

E -

F - 7:40

G -

Room 101

Room 102

Room 103

VIDEO SOLUTION

Answers

1. 200	2. 2	3. B	4. D	5. A	6. A	7. 0	8. B
9. 2	10. D	11. B	12. B	13. A	14. C	15. C	16. A
17. D	18. D	19. C	20. A	21. B	22. C	23. D	24. B
25. D	26. C	27. B	28. A	29. D	30. A	31. D	32. 8
33. C	34. A						

Explanations

1. 200

It is given that the applications are scheduled for processing in twenty 15-minute slots starting at 9:00 am and ending at 2:00 pm. Ten applications are scheduled in each slot.

Hence, the total number of applicants = $(20 \times 10) = 200$. It is also known that 50% of the applications were US applications, and the number of US applications was the same in all the slots. The same was true for the other three categories.

Hence, the number of total number of US applicants = $(200 \times 50\%) = 100$, and the number of US applicants in each slot = $(100/20) = 5$

It is also known that Ira, Vijay, and Nandini were scheduled for Schengen visa processing in that order. They had a 9:15 am slot. Since the number of Schengen applicants was the same in all the slots, it implies the number of Schengen applicants in each slot is at least 3.

Similarly, it is given that Mahira and Osman were scheduled in the 9:30 am slot on that day for visa processing in the Others category, which implies the number of other category applicants in each slot is at least 2. Since the number of total applicants in each slot is 10, this implies the number of Schengen and other applicants in each slot is 3, and 2, respectively. Hence, the number of UK applicants is 0 in each slot.

It is also known that the number of total counters is 10, among which four are dedicated to US applications, and two each for UK applications, Schengen applications, and Others applications. It is given that each US and UK application requires 10 minutes of processing time, and Vijay was called to a counter at 9:25 am. (Who is 5th in the queue). It can only be possible when the processing time of Schengen applications is 12.5 minutes.

US (10 min)				Schengen (12.5 min)		Others (5 min process)	
End Time				End Time		End Time	
C1	C2	C3	C4	C1	C2	C1	C2
9.10	9.10	9.10	9.10	9.12.30	9.12.30	9.05	9.05
9.20	9.25	9.25	9.25	9.25	9.32.30	9.20	9.20
9.30	9.35	9.40	9.40	9.37.30	9.45	9.35	9.35
9.40	9.45	9.50	9.55				
9.55	9.55	10.00	10.05				
10.10	10.10	10.10	10.15				
10.20							

On a particular day, Ira, Vijay, and Nandini were scheduled for Schengen visa processing in that order. They had a 9:15 am slot but entered the VPO at 9:20 am. When they entered the office, exactly six out of the ten counters were either processing applications, or had finished processing one and ready to start processing the next. Hence, at 9:20 am, there are exactly four free counters. Out of these 4, 2 is the UK counter, and the other two are other counters. (Since the US counters and Schengen Counters were either processing applications, or had finished processing one and were ready to start processing the next.)

For the other applicants, the time taken to process one application is at most 5 minutes, which implies the total time taken to process 40 applications is at most $(40 \times 5) = 200$ minutes.

VIDEO SOLUTION

2.2

It is given that none of the streets has more than one team traveling along it in any direction at any point in time (point 1), which implies at 9.00 hrs, all 4 teams have chosen different roots from the starting point.

It is also known that Teams 2 and 3 are the only ones in stations E and D respectively at 10:00 hrs, and Team 1 and Team 4 are the only teams that patrol the street connecting stations A and E.

It is only possible when Team 2 traveled (A-E) via F, and Team 3 reached station D via station C.

It is also known that Teams 1 and 3 are the only ones in Station E at 10:30 hrs, and Team 4 never passes through Stations B, D, or F. Hence, Team 1 must have chosen the (A-B) root at the starting point, and Team 4 has chosen the (A-E) root at 9.00 hrs.

Hence, Team 1 will reach B at 9.30, and come to A at 10.00 hrs. After that, they will go to E at 10.30 hrs.

Since Team 4 never passes through stations B, D, or F. Team 4 only can pass through stations A, E, and C.

Hence, the roots of team 4 to reach station E at 11.30 will be (A-E-A-C-A-E) or (A-E-A-E-A-E).

Since team 1 is already traveling to E from A at 10.00 hrs, at that time team 4 can't choose the same route.

Hence, the final route for team 4 to reach E at 11.30 is (A-E-A-C-A-E), and at 12.00 hrs, team 4 will come back to station A.

Hence, the complete route diagram for team 4 is (A-E-A-C-A-E-A)

Teams	9.00	9.30	10.00	10.30	11.00	11.30	12.00
1	A	B	A	E			
2	A	F	E				
3	A	C	D				
4	A	E	A	C	A	E	A

We can see that team 1 is at station E at 10.30 hrs, and they will reach station B at 11.30 hrs, which is only possible when they travel to B via A.

Hence, the complete route diagram for team 1 is (A-B-A-E-A-B-A). It is also known that Teams 1 and 3 are the only ones in station E at 10:30 hrs.

Teams	9.00	9.30	10.00	10.30	11.00	11.30	12.00
1	A	B	A	E	A	B	A
2	A	F	E				
3	A	C	D	E			
4	A	E	A	C	A	E	A

The only possible root for Team 2 at 10.00 hrs is from E to F since they can't choose E to D because Team 3 is already on this route. Since team 3 has to reach A at 12.00. The only possible combination for team 3 is E-D-C-A

Teams	9.00	9.30	10.00	10.30	11.00	11.30	12.00
1	A	B	A	E	A	B	A
2	A	F	E	F			
3	A	C	D	E	D	C	A
4	A	E	A	C	A	E	A

Now the roots for team 2 going back to A is from F at 10.30 hrs (F-A-F-A) or (F-E-F-A).

Hence, the final table is given below:

Teams	9.00	9.30	10.00	10.30	11.00	11.30	12.00
1	A	B	A	E	A	B	A
2	A	F	E	F	A/E	F	A
3	A	C	D	E	D	C	A
4	A	E	A	C	A	E	A

From the table, we can see that the teams have passed through B 2 times in this given period.

[VIDEO SOLUTION](#)

3. B

P (2)	Sociology	B, C
Q (1)	Anthropology/ Sociology	Either F or G
R (2)	Economics	A, Either of D/ E/ H
S (1)	Sociology/ Anthropology	G/ F
T (2)	Economics	Any two of the D/ E/ H

09:00			09:30			10:00			10:30		
Teacher	Student	Subject	Teacher	Student	Subject	Teacher	Student	Subject	Teacher	Student	Subject
T		Economics	T		Economics	R	A	Economics	R		Economics
		Sociology		B	Sociology					F	Anthropology
				G	Sociology						

Now D is scheduled in the slot immediately before Q. In that case Q will be in the 9:30 slot and D will be in the 9:00 am slot. Hence D will be mentored by T and one of E/H will be mentored by R and the other by T. Obviously E is guided by R is not necessarily true as he can be mentored by T also.

[VIDEO SOLUTION](#)

4. D

Let's look at all the students we have A,B,C,D,E,F,G,H. The guides are P,Q,R,S,T and they mentor 2,1,2,1,2 students respectively.

Slots are 9:00,9:30,10:00,10:30 . Moreover economics, sociology and anthropology have 4,3,1 students respectively.

Now from 5th statement, two sociology seminars and economics seminars are held in the same slot. This will not take place at 10:00(only one seminar is scheduled at 10:00), this will also not take place at 10:30(An anthropology seminar is scheduled at 10:30 and maximum three seminars are scheduled in a slot. Hence this will happen at either 9 or 9:30. Now from (7) A and G are scheduled in consecutive slots. A is scheduled at 10:00. Hence G will be scheduled at 9:30. SO from (5), there will be three students having their seminars at 9:30.

Combining this data with the remaining points we get the following table

P (2)	Sociology	B, C
Q (1)	Anthropology/ Sociology	Either F or G
R (2)	Economics	<u>A</u> , Either of D/ E/ H
S (1)	Sociology/ Anthropology	G/ F
T (2)	Economics	Any two of the D/ E/ H

Moreover we do know that students having same subject are scheduled in consecutive slots. So we get

09:00			09:30			10:00			10:30		
Teacher	Student	Subject	Teacher	Student	Subject	Teacher	Student	Subject	Teacher	Student	Subject
T		Economics	T		Economics	R	A	Economics	R		Economics
		Sociology		B	Sociology					F	Anthropology
				G	Sociology						

As we can see from the above table, P,Q and S are not guiding any economics student

[VIDEO SOLUTION](#)

5.A

Let's look at all the students we have A,B,C,D,E,F,G,H. The guides are P,Q,R,S,T and they mentor 2,1,2,1,2 students respectively.

Slots are 9:00,9:30,10:00,10:30 . Moreover economics, sociology and anthropology have 4,3,1 students respectively.

Now from 5th statement, two sociology seminars and economics seminars are held in the same slot. This will not take place at 10:00(only one seminar is scheduled at 10:00), this will also not take place at 10:30(An anthropology seminar is scheduled at 10:30 and maximum three seminars are scheduled in a slot. Hence this will happen at either 9 or 9:30. Now from (7) A and G are scheduled in consecutive slots. A is scheduled at 10:00. Hence G will be scheduled at 9:30. SO from (5), there will be three students having their seminars at 9:30.

Combining this data with the remaining points we get the following table

P (2)	Sociology	B, C
Q (1)	Anthropology/ Sociology	Either F or G
R (2)	Economics	<u>A</u> , Either of D/ E/ H
S (1)	Sociology/ Anthropology	G/ F
T (2)	Economics	Any two of the D/ E/ H

Moreover we do know that students having same subject are scheduled in consecutive slots. So we get

09:00			09:30			10:00			10:30		
Teacher	Student	Subject	Teacher	Student	Subject	Teacher	Student	Subject	Teacher	Student	Subject
T		Economics	T		Economics	R	A	Economics	R		Economics
		Sociology		B	Sociology					F	Anthropology
				G	Sociology						

Here we can see that two seminars are scheduled in first slot.

[VIDEO SOLUTION](#)

6. A

It is given that the applications are scheduled for processing in twenty 15-minute slots starting at 9:00 am and ending at 2:00 pm. Ten applications are scheduled in each slot.

Hence, the total number of applicants = $(20 \times 10) = 200$. It is also known that 50% of the applications were US applications, and the number of US applications was the same in all the slots. The same was true for the other three categories.

Hence, the number of total number of US applicants = $(200 \times 50\%) = 100$, and the number of US applicants in each slot = $(100/20) = 5$

It is also known that Ira, Vijay, and Nandini were scheduled for Schengen visa processing in that order. They had a 9:15 am slot. Since the number of Schengen applicants was the same in all the slots, it implies the number of Schengen applicants in each slot is at least 3.

Similarly, it is given that Mahira and Osman were scheduled in the 9:30 am slot on that day for visa processing in the Others category, which implies the number of other category applicants in each slot is at least 2. Since the number of total applicants in each slot is 10, this implies the number of Schengen and other applicants in each slot is 3, and 2, respectively. Hence, the number of UK applicants is 0 in each slot.

It is also known that the number of total counters is 10, among which four are dedicated to US applications, and two each for UK applications, Schengen applications, and Others applications. It is given that each US and UK application requires 10 minutes of processing time, and Vijay was called to a counter at 9:25 am. (Who is 5th in the queue). It can only be possible when the processing time of Schengen applications is 12.5 minutes.

US (10 min)				Schengen (12.5 min)		Others (5 min process)	
End Time				End Time		End Time	
C1	C2	C3	C4	C1	C2	C1	C2
9.10	9.10	9.10	9.10	9.12.30	9.12.30	9.05	9.05
9.20	9.25	9.25	9.25	9.25	9.32.30	9.20	9.20
9.30	9.35	9.40	9.40	9.37.30	9.45	9.35	9.35
9.40	9.45	9.50	9.55				
9.55	9.55	10.00	10.05				
10.10	10.10	10.10	10.15				
10.20							

On a particular day, Ira, Vijay, and Nandini were scheduled for Schengen visa processing in that order. They had a 9:15 am slot but entered the VPO at 9:20 am. When they entered the office, exactly six out of the ten counters were either processing applications, or had finished processing one and ready to start processing the next. Hence, at 9:20 am, there are exactly four free counters. Out of these 4, 2 is the UK counter, and the other two are other counters. (Since the US counters and Schengen Counters were either processing applications, or had finished processing one and were ready to start processing the next.)

From the table, we can see that the first slot takes 20 minutes to complete, and after that remaining 19 slots take 15 minutes each to complete the US application process.

Hence, the total time taken = $20 + 15 \times 19 = 305$ minutes = 5 hrs 5 minutes. Hence, the time will be (9 am + 5hrs 5 minutes) = 2.05 pm

The correct option is A

 VIDEO SOLUTION

7. 0

It is given that the applications are scheduled for processing in twenty 15-minute slots starting at 9:00 am and ending at 2:00 pm. Ten applications are scheduled in each slot.

Hence, the total number of applicants = $(20 \times 10) = 200$. It is also known that 50% of the applications were US applications, and the number of US applications was the same in all the slots. The same was true for the other three categories.

Hence, the number of total number of US applicants = $(200 \times 50\%) = 100$, and the number of US applicants in each slot = $(100/20) = 5$

It is also known that Ira, Vijay, and Nandini were scheduled for Schengen visa processing in that order. They had a 9:15 am slot. Since the number of Schengen applicants was the same in all the slots, it implies the number of Schengen applicants in each slot is at least 3.

Similarly, it is given that Mahira and Osman were scheduled in the 9:30 am slot on that day for visa processing in the Others category, which implies the number of other category applicants in each slot is at least 2. Since the number of total applicants in each slot is 10, this implies the number of Schengen and other applicants in each slot is 3, and 2, respectively. Hence, the number of UK applicants is 0 in each slot.

It is also known that the number of total counters is 10, among which four are dedicated to US applications, and two each for UK applications, Schengen applications, and Others applications. It is given that each US and UK application requires 10 minutes of processing time, and Vijay was called to a counter at 9:25 am. (Who is 5th in the queue). It can only be possible when the processing time of Schengen applications is 12.5 minutes.

US (10 min)				Schengen (12.5 min)		Others (5 min process)	
End Time				End Time		End Time	
C1	C2	C3	C4	C1	C2	C1	C2
9.10	9.10	9.10	9.10	9.12.30	9.12.30	9.05	9.05
9.20	9.25	9.25	9.25	9.25	9.32.30	9.20	9.20
9.30	9.35	9.40	9.40	9.37.30	9.45	9.35	9.35
9.40	9.45	9.50	9.55				
9.55	9.55	10.00	10.05				
10.10	10.10	10.10	10.15				
10.20							

On a particular day, Ira, Vijay, and Nandini were scheduled for Schengen visa processing in that order. They had a 9:15 am slot but entered the VPO at 9:20 am. When they entered the office, exactly six out of the ten counters were either processing applications, or had finished processing one and ready to start processing the next. Hence, at 9:20 am, there are exactly four free counters. Out of these 4, 2 is the UK counter, and the other two are other counters. (Since the US counters and Schengen Counters were either processing applications, or had finished processing one and were ready to start processing the next.)

From the table, we can say that the total number of UK applicants in each slot is zero, Hence, the total number of applicants is zero.

 VIDEO SOLUTION

8. B

It is given that the applications are scheduled for processing in twenty 15-minute slots starting at 9:00 am and ending at 2:00 pm. Ten applications are scheduled in each slot.

Hence, the total number of applicants = $(20 \times 10) = 200$. It is also known that 50% of the applications were US applications, and the number of US applications was the same in all the slots. The same was true for the other three categories.

Hence, the number of total number of US applicants = $(200 \times 50\%) = 100$, and the number of US applicants in each slot = $(100/20) = 5$

It is also known that Ira, Vijay, and Nandini were scheduled for Schengen visa processing in that order. They had a 9:15 am slot. Since the number of Schengen applicants was the same in all the slots, it implies the number of Schengen applicants in each slot is at least 3.

Similarly, it is given that Mahira and Osman were scheduled in the 9:30 am slot on that day for visa processing in the Others category, which implies the number of other category applicants in each slot is at least 2. Since the number of total applicants in each slot is 10, this implies the number of Schengen and other applicants in each slot is 3, and 2, respectively. Hence, the number of UK applicants is 0 in each slot.

It is also known that the number of total counters is 10, among which four are dedicated to US applications, and two each for UK applications, Schengen applications, and Others applications. It is given that each US and UK application requires 10 minutes of processing time, and Vijay was called to a counter at 9:25 am. (Who is 5th in the queue). It can only be possible when the processing time of Schengen applications is 12.5 minutes.

US (10 min)				Schengen (12.5 min)		Others (5 min process)	
End Time				End Time		End Time	
C1	C2	C3	C4	C1	C2	C1	C2
9.10	9.10	9.10	9.10	9.12.30	9.12.30	9.05	9.05
9.20	9.25	9.25	9.25	9.25	9.32.30	9.20	9.20
9.30	9.35	9.40	9.40	9.37.30	9.45	9.35	9.35
9.40	9.45	9.50	9.55				
9.55	9.55	10.00	10.05				
10.10	10.10	10.10	10.15				
10.20							

On a particular day, Ira, Vijay, and Nandini were scheduled for Schengen visa processing in that order. They had a 9:15 am slot but entered the VPO at 9:20 am. When they entered the office, exactly six out of the ten counters were either processing applications, or had finished processing one and ready to start processing the next. Hence, at 9:20 am, there are exactly four free counters. Out of these 4, 2 is the UK counter, and the other two are other counters. (Since the US counters and Schengen Counters were either processing applications, or had finished processing one and were ready to start processing the next.)

Let's check the options.

Option A: The application process of Osman was completed before 9:45 am. => True (Since he is 5th in the queue, his process will end at 9:35 am)

Option B: The application process of Mahira started after Nandini's. => The application process for Mahira starts at 9:30 am, and the application process for Nandini starts at 9:32.30 am => False.

The correct option is B

VIDEO SOLUTION

9.2

It is given that none of the streets has more than one team traveling along it in any direction at any point in time (point 1), which implies at 9.00 hrs, all 4 teams have chosen different roots from the starting point.

It is also known that Teams 2 and 3 are the only ones in stations E and D respectively at 10:00 hrs, and Team 1 and Team 4 are the only teams that patrol the street connecting stations A and E.

It is only possible when Team 2 traveled (A-E) via F, and Team 3 reached station D via station C.

It is also known that Teams 1 and 3 are the only ones in Station E at 10:30 hrs, and Team 4 never passes through Stations B, D, or F. Hence, Team 1 must have chosen the (A-B) root at the starting point, and Team 4 has chosen the (A-E) root at 9.00 hrs.

Hence, Team 1 will reach B at 9.30, and come to A at 10.00 hrs. After that, they will go to E at 10.30 hrs.

Since Team 4 never passes through stations B, D, or F. Team 4 only can pass through stations A, E, and C.

Hence, the roots of team 4 to reach station E at 11.30 will be (A-E-A-C-A-E) or (A-E-A-E-A-E).

Since team 1 is already traveling to E from A at 10.00 hrs, at that time team 4 can't choose the same route. Hence, the final route for team 4 to reach E at 11.30 is (A-E-A-C-A-E), and at 12.00 hrs, team 4 will come back to station A.

Hence, the complete route diagram for team 4 is (A-E-A-C-A-E-A)

Teams	9.00	9.30	10.00	10.30	11.00	11.30	12.00
1	A	B	A	E			
2	A	F	E				
3	A	C	D				
4	A	E	A	C	A	E	A

We can see that team 1 is at station E at 10.30 hrs, and they will reach station B at 11.30 hrs, which is only possible when they travel to B via A.

Hence, the complete route diagram for team 1 is (A-B-A-E-A-B-A). It is also known that Teams 1 and 3 are the only ones in station E at 10:30 hrs.

Teams	9.00	9.30	10.00	10.30	11.00	11.30	12.00
1	A	B	A	E	A	B	A
2	A	F	E				
3	A	C	D	E			
4	A	E	A	C	A	E	A

The only possible route for Team 2 at 10.00 hrs is from E to F since they can't choose E to D because Team 3 is already on this route. Since team 3 has to reach A at 12.00. The only possible combination for team 3 is E-D-C-A

Teams	9.00	9.30	10.00	10.30	11.00	11.30	12.00
1	A	B	A	E	A	B	A
2	A	F	E	F			
3	A	C	D	E	D	C	A
4	A	E	A	C	A	E	A

Now the route for team 2 going back to A is from F at 10.30 hrs (F-A-F-A) or (F-E-F-A).

Hence, the final table is given below:

Teams	9.00	9.30	10.00	10.30	11.00	11.30	12.00
1	A	B	A	E	A	B	A
2	A	F	E	F	A/E	F	A
3	A	C	D	E	D	C	A
4	A	E	A	C	A	E	A

From the table, we can see that team 4 passed station E 2 times in a day

[VIDEO SOLUTION](#)

10.D

It is given that none of the streets has more than one team traveling along it in any direction at any point in time (point 1), which implies at 9.00 hrs, all 4 teams have chosen different roots from the starting point.

It is also known that Teams 2 and 3 are the only ones in stations E and D respectively at 10:00 hrs, and Team 1 and Team 4 are the only teams that patrol the street connecting stations A and E.

It is only possible when Team 2 traveled (A-E) via F, and Team 3 reached station D via station C.

It is also known that Teams 1 and 3 are the only ones in Station E at 10:30 hrs, and Team 4 never passes through Stations B, D, or F. Hence, Team 1 must have chosen the (A-B) root at the starting point, and Team 4 has chosen the (A-E) root at 9.00 hrs.

Hence, Team 1 will reach B at 9.30, and come to A at 10.00 hrs. After that, they will go to E at 10.30 hrs.

Since Team 4 never passes through stations B, D, or F. Team 4 only can pass through stations A, E, and C.

Hence, the roots of team 4 to reach station E at 11.30 will be (A-E-A-C-A-E) or (A-E-A-E-A-E).

Since team 1 is already traveling to E from A at 10.00 hrs, at that time team 4 can't choose the same route.

Hence, the final route for team 4 to reach E at 11.30 is (A-E-A-C-A-E), and at 12.00 hrs, team 4 will come back to station A.

Hence, the complete route diagram for team 4 is (A-E-A-C-A-E-A)

Teams	9.00	9.30	10.00	10.30	11.00	11.30	12.00
1	A	B	A	E			
2	A	F	E				
3	A	C	D				
4	A	E	A	C	A	E	A

We can see that team 1 is at station E at 10.30 hrs, and they will reach station B at 11.30 hrs, which is only possible when they travel to B via A.

Hence, the complete route diagram for team 1 is (A-B-A-E-A-B-A). It is also known that Teams 1 and 3 are the only ones in station E at 10:30 hrs.

Teams	9.00	9.30	10.00	10.30	11.00	11.30	12.00
1	A	B	A	E	A	B	A
2	A	F	E				
3	A	C	D	E			
4	A	E	A	C	A	E	A

The only possible root for Team 2 at 10.00 hrs is from E to F since they can't choose E to D because Team 3 is already on this route. Since team 3 has to reach A at 12.00. The only possible combination for team 3 is E-D-C-A

Teams	9.00	9.30	10.00	10.30	11.00	11.30	12.00
1	A	B	A	E	A	B	A
2	A	F	E	F			
3	A	C	D	E	D	C	A
4	A	E	A	C	A	E	A

Now the roots for team 2 going back to A is from F at 10.30 hrs (F-A-F-A) or (F-E-F-A).

Hence, the final table is given below:

Teams	9.00	9.30	10.00	10.30	11.00	11.30	12.00
1	A	B	A	E	A	B	A
2	A	F	E	F	A/E	F	A
3	A	C	D	E	D	C	A
4	A	E	A	C	A	E	A

From the table, we can see that 2 teams (teams 3 and 4) have passed through station C on the given day.

The correct option is D

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11.B

It is given that

1 burger takes 10 minutes

1 ice cream takes 2 minutes and 3 portions of fries take 5 min by the machine (operator is not required).

Client 1 ordered 1 burger, 3 portions of fries and 1 ice cream.

The burger will take 10 minutes to finish. Meantime, rest of the order can be completed.

12. B

It is given that

1 burger takes 10 minutes

1 ice cream takes 2 minutes and 3 portions of fries take 5 min by the machine (operator is not required).

The employees are allowed to process multiple orders at a time, but the preference would be to finish orders of clients who placed their orders earlier.

The first order is 1 burger, 1 ice cream and 3 portions of fries.

Anish can start working on the burger and Bani can start working on the ice cream.

The burger will be done at 10:10, ice - cream at 10:02 and fries at 10:05.

The second order is placed at 10:05. (ice cream and fries)

Bani can work on the ice cream from 10:05 and also put the fries.

Only Bani will be free from 10:02 to 10:05 - 3 minutes.

The ice cream will be done by 10:07 but the fries will be done by 10:10.

Third order of 1 burger and 1 fries is placed at 10:07.

Bani will immediately start working on the burger. He will finish it by 10:17.

Anish would have finished the first burger at 10:10. Only he is free till 10:17 - 7 minutes.

Thus exactly one person is free for 10 minutes.

13. A

In the information, it is provided that there are four authors in the logistics department, 2 authors in automation, and 2 authors in manufacturing. There are 4 Indian authors, 2 Chinese authors, and 2 Japanese authors.

The total issues publicized were 18 of which the publications done by A, D, G, H are twice the papers written by the other four authors. Hence $A+D+G+H$ has written 12 papers in total and $B+C+E+F$ has written 6 in total. Since an author can write a minimum of 4 papers and a maximum of 3 papers. Each of A, D, G, H must have written 3 papers each.

In statement 3 it was provided that none of the Indian authors were from the Manufacturing area and none of the Japanese or Chinese authors were from the Automation area.

Since none of the Japanese and Chinese authors belonged to automation. Hence the two automation authors must be from India. In statement 1 it was given that F an Indian author is from logistics.

Of the remaining 5 authors 3 from logistics and 2 from manufacturing, in statement 5 it is given that the two Chinese authors are from different areas and hence they must be from manufacturing and logistics.

Of the remaining 3 authors 2 in logistics and 1 in manufacturing, in statement 8 is given that a Japanese author belonged to manufacturing, and hence of the remaining 2 in logistics one of them was from India and the other from Japan.

Of the four Indian authors, 2 belonged to logistics and 2 automation. Of the two authors from Japan, one belonged to manufacturing and one logistics. Of the two Chinese authors, one of them belonged to logistics and the other manufacturing.

Five papers were scheduled in each of the January and April issues, while four were scheduled in each of July and October issues.

Using statement 1, F an Indian author wrote a single paper in logistics published in October.

Using statement 2, A was from Automation and did not have a paper scheduled in October. Since A wrote 3 papers in total he must have written in the other three months and since only Indian authors worked in Automation he must have been from India.

Using statement 5, C, E are Chinese and wrote an equal number of papers. Since $B+C+E+F$ have written a total of 6 papers. The two possibilities are $2+2+1+1$ or $3+1+1+1$.. Since it was given that E had papers scheduled in consecutive issues and C did not. So the only possible case is $C = 2, E = 2, F = 1, B = 1$.

Using statement 4, A and H were from different countries and wrote papers in the same months. Hence H must be Japanese and must have written in Jan, April, July.

Author	Country	Area of Interest	Publications	Months of publication
A	India	Automation	3	Jan, April, July
B			1	
C	Chinese		2	
D			3	
E	Chinese		2	
F	India	Logistics	1	October
G			3	
H	Japan		3	Jan, April, July

In statement 6, B, from the Logistics area, had a paper scheduled in the April issue of the journal.

In statement 7, B and G belonged to the same country. None of their papers were scheduled in the same issue of the journal. Since B and G belonged to the same country the only possibility is that both of them belonged to the same country and since they published Journals in different months. G must have published in January, July, and October.

In statement 8, D, a Japanese author from the Manufacturing area, did not have a paper scheduled in the July issue. Hence he must have had the three issues in Jan, April, Oct. The other Japanese author H must have written in logistics.

The fourth Indian author G must have written a paper in Automation.

In January, one more paper needs to be publicized, one in April, 1 in July, and one in October.

In statement 5, E had papers scheduled in consecutive issues of the journal but C did not.

Hence C must have written in Jan and October, and E in April and July.

In statement 9, C and H belonged to different areas. Hence C must be from Manufacturing and E must be from Logistics.

Author	Country	Area of Interest	Publications	Months of publication
A	India	Automation	3	Jan, April, July
B	India	Logistics	1	April
C	China	Manufacturing	2	Jan, October
D	Japan	Manufacturing	3	Jan, April, October
E	China	Logistics	2	April, July.
F	India	Logistics	1	October
G	India	Automation	3	Jan, July, October
H	Japan	Logistics	3	Jan, April, July

.Number of papers written by B, C, E, and G are: 1, 2, 2, 3

 VIDEO SOLUTION

14. C

In the information, it is provided that there are four authors in the logistics department, 2 authors in automation, and 2 authors in manufacturing. There are 4 Indian authors, 2 Chinese authors, and 2 Japanese authors.

The total issues publicized were 18 of which the publications done by A, D, G, H are twice the papers written by the other four authors. Hence $A+D+G+H$ has written 12 papers in total and $B+C+E+F$ has written 6 in total. Since an author can write a minimum of 4 papers and a maximum of 3 papers. Each of A, D, G, H must have written 3 papers each.

In statement 3 it was provided that none of the Indian authors were from the Manufacturing area and none of the Japanese or Chinese authors were from the Automation area.

Since none of the Japanese and Chinese authors belonged to automation. Hence the two automation authors must be from India. In statement 1 it was given that F an Indian author is from logistics.

Of the remaining 5 authors 3 from logistics and 2 from manufacturing, in statement 5 it is given that the two Chinese authors are from different areas and hence they must be from manufacturing and logistics.

Of the remaining 3 authors 2 in logistics and 1 in manufacturing, in statement 8 is given that a Japanese author belonged to manufacturing, and hence of the remaining 2 in logistics one of them was from India and the other from Japan.

Of the four Indian authors, 2 belonged to logistics and 2 automation. Of the two authors from Japan, one belonged to manufacturing and one logistics. Of the two Chinese authors, one of them belonged to logistics and the other manufacturing.

Five papers were scheduled in each of the January and April issues, while four were scheduled in each of July and October issues.

Using statement 1, F an Indian author wrote a single paper in logistics published in October.

Using statement 2, A was from Automation and did not have a paper scheduled in October. Since A wrote 3 papers in total he must have written in the other three months and since only Indian authors worked in Automation he must have been from India.

Using statement 5, C, E are Chinese and wrote an equal number of papers. Since $B+C+E+F$ have written a total of 6 papers. The two possibilities are $2+2+1+1$ or $3+1+1+1$. Since it was given that E had papers scheduled in consecutive issues and C did not. So the only possible case is $C = 2, E = 2, F = 1, B = 1$.

Using statement 4, A and H were from different countries and wrote papers in the same months. Hence H must be Japanese and must have written in Jan, April, July.

Author	Country	Area of Interest	Publications	Months of publication
A	India	Automation	3	Jan, April, July
B			1	
C	Chinese		2	
D			3	
E	Chinese		2	
F	India	Logistics	1	October
G			3	
H	Japan		3	Jan, April, July

In statement 6, B, from the Logistics area, had a paper scheduled in the April issue of the journal.

In statement 7, B and G belonged to the same country. None of their papers were scheduled in the same issue of the journal. Since B and G belonged to the same country the only possibility is that both of them belonged to the same country and since they published Journals in different months. G must have published in January, July, and October.

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The fourth Indian author G must have written a paper in Automation.

In January, one more paper needs to be publicized, one in April, 1 in July, and one in October.

In statement 5, E had papers scheduled in consecutive issues of the journal but C did not.

Hence C must have written in Jan and October, and E in April and July.

In statement 9, C and H belonged to different areas. Hence C must be from Manufacturing and E must be from Logistics.

Author	Country	Area of Interest	Publications	Months of publication
A	India	Automation	3	Jan, April, July
B	India	Logistics	1	April
C	China	Manufacturing	2	Jan, October
D	Japan	Manufacturing	3	Jan, April, October
E	China	Logistics	2	April, July.
F	India	Logistics	1	October
G	India	Automation	3	Jan, July, October
H	Japan	Logistics	3	Jan, April, July

The publication of April has three authors from logistics. Hence false

 VIDEO SOLUTION

15. C

In the information, it is provided that there are four authors in the logistics department, 2 authors in automation, and 2 authors in manufacturing. There are 4 Indian authors, 2 Chinese authors, and 2 Japanese authors.

The total issues publicized were 18 of which the publications done by A, D, G, H are twice the papers written by the other four authors. Hence A+D+G+H has written 12 papers in total and B+C+E+F has written 6 in total. Since an author can write a minimum of 4 papers and a maximum of 3 papers. Each of A, D, G, H must have written 3 papers each.

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Of the remaining 5 authors 3 from logistics and 2 from manufacturing, in statement 5 it is given that the two Chinese authors are from different areas and hence they must be from manufacturing and logistics.

Of the remaining 3 authors 2 in logistics and 1 in manufacturing, in statement 8 is given that a Japanese author belonged to manufacturing, and hence of the remaining 2 in logistics one of them was from India and the other from Japan.

Of the four Indian authors, 2 belonged to logistics and 2 automation. Of the two authors from Japan, one belonged to manufacturing and one logistics. Of the two Chinese authors, one of them belonged to logistics and the other manufacturing.

Five papers were scheduled in each of the January and April issues, while four were scheduled in each of July and October issues.

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Using statement 5, C, E are Chinese and wrote an equal number of papers. Since B+C+E+F have written a total of 6 papers. The two possibilities are 2+2+1+1 or 3+1+1+1.. Since it was given that E had papers scheduled in consecutive issues and C did not. So the only possible case is C = 2, E = 2, F = 1, B = 1.

Using statement 4, A and H were from different countries and wrote papers in the same months. Hence H must be Japanese and must have written in Jan, April, July.

Author	Country	Area of Interest	Publications	Months of publication
A	India	Automation	3	Jan, April, July
B			1	
C	Chinese		2	
D			3	
E	Chinese		2	
F	India	Logistics	1	October
G			3	
H	Japan		3	Jan, April, July

In statement 6, B, from the Logistics area, had a paper scheduled in the April issue of the journal.

In statement 7, B and G belonged to the same country. None of their papers were scheduled in the same issue of the journal. Since B and G belonged to the same country the only possibility is that both of them belonged to the same country and since they published Journals in different months. G must have published in January, July, and October.

In statement 8, D, a Japanese author from the Manufacturing area, did not have a paper scheduled in the July issue. Hence he must have had the three issues in Jan, April, Oct. The other Japanese author H must have written in logistics.

The fourth Indian author G must have written a paper in Automation.

In January, one more paper needs to be publicized, one in April, 1 in July, and one in October.

In statement 5, E had papers scheduled in consecutive issues of the journal but C did not.

Hence C must have written in Jan and October, and E in April and July.

In statement 9, C and H belonged to different areas. Hence C must be from Manufacturing and E must be from Logistics.

Author	Country	Area of Interest	Publications	Months of publication
A	India	Automation	3	Jan, April, July
B	India	Logistics	1	April
C	China	Manufacturing	2	Jan, October
D	Japan	Manufacturing	3	Jan, April, October
E	China	Logistics	2	April, July.
F	India	Logistics	1	October
G	India	Automation	3	Jan, July, October
H	Japan	Logistics	3	Jan, April, July

.Option C is false in the issue of April there were three authors from logistics department.

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16. A

In the information, it is provided that there are four authors in the logistics department, 2 authors in automation, and 2 authors in manufacturing. There are 4 Indian authors, 2 Chinese authors, and 2 Japanese authors.

The total issues publicized were 18 of which the publications done by A, D, G, H are twice the papers written by the other four authors. Hence $A+D+G+H$ has written 12 papers in total and $B+C+E+F$ has written 6 in total. Since an author can write a minimum of 4 papers and a maximum of 3 papers. Each of A, D, G, H must have written 3 papers each.

In statement 3 it was provided that none of the Indian authors were from the Manufacturing area and none of the Japanese or Chinese authors were from the Automation area.

Since none of the Japanese and Chinese authors belonged to automation. Hence the two automation authors must be from India. In statement 1 it was given that F an Indian author is from logistics.

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Of the remaining 3 authors 2 in logistics and 1 in manufacturing, in statement 8 is given that a Japanese author belonged to manufacturing, and hence of the remaining 2 in logistics one of them was from India and the other from Japan.

Of the four Indian authors, 2 belonged to logistics and 2 automation. Of the two authors from Japan, one belonged to manufacturing and one logistics. Of the two Chinese authors, one of them belonged to logistics and the other manufacturing.

Five papers were scheduled in each of the January and April issues, while four were scheduled in each of July and October issues.

Using statement 1, F an Indian author wrote a single paper in logistics published in October.

Using statement 2, A was from Automation and did not have a paper scheduled in October. Since A wrote 3 papers in total he must have written in the other three months and since only Indian authors worked in Automation he must have been from India.

Using statement 5, C, E are Chinese and wrote an equal number of papers. Since B+C+E+F have written a total of 6 papers. The two possibilities are 2+2+1+1 or 3+1+1+1.. Since it was given that E had papers scheduled in consecutive issues and C did not. So the only possible case is C = 2, E = 2, F = 1, B = 1.

Using statement 4, A and H were from different countries and wrote papers in the same months. Hence H must be Japanese and must have written in Jan, April, July.

Author	Country	Area of Interest	Publications	Months of publication
A	India	Automation	3	Jan, April, July
B			1	
C	Chinese		2	
D			3	
E	Chinese		2	
F	India	Logistics	1	October
G			3	
H	Japan		3	Jan, April, July

In statement 6, B, from the Logistics area, had a paper scheduled in the April issue of the journal.

In statement 7, B and G belonged to the same country. None of their papers were scheduled in the same issue of the journal. Since B and G belonged to the same country the only possibility is that both of them belonged to the same country and since they published Journals in different months. G must have published in January, July, and October.

In statement 8, D, a Japanese author from the Manufacturing area, did not have a paper scheduled in the July issue. Hence he must have had the three issues in Jan, April, Oct. The other Japanese author H must have written in logistics.

The fourth Indian author G must have written a paper in Automation.

In January, one more paper needs to be publicized, one in April, 1 in July, and one in October.

In statement 5, E had papers scheduled in consecutive issues of the journal but C did not.

Hence C must have written in Jan and October, and E in April and July.

In statement 9, C and H belonged to different areas. Hence C must be from Manufacturing and E must be from Logistics.

Author	Country	Area of Interest	Publications	Months of publication
A	India	Automation	3	Jan, April, July
B	India	Logistics	1	April
C	China	Manufacturing	2	Jan, October
D	Japan	Manufacturing	3	Jan, April, October
E	China	Logistics	2	April, July.
F	India	Logistics	1	October
G	India	Automation	3	Jan, July, October
H	Japan	Logistics	3	Jan, April, July

In Automation, there are a total of 6 papers, Logistics 7 papers, and Manufacturing a total of 5 papers.

VIDEO SOLUTION

17. D

It is given that none of the streets has more than one team traveling along it in any direction at any point in time (point 1), which implies at 9.00 hrs, all 4 teams have chosen different roots from the starting point.

It is also known that Teams 2 and 3 are the only ones in stations E and D respectively at 10:00 hrs, and Team 1 and Team 4 are the only teams that patrol the street connecting stations A and E.

It is only possible when Team 2 traveled (A-E) via F, and Team 3 reached station D via station C.

It is also known that Teams 1 and 3 are the only ones in Station E at 10:30 hrs, and Team 4 never passes through Stations B, D, or F. Hence, Team 1 must have chosen the (A-B) root at the starting point, and Team 4 has chosen the (A-E) root at 9.00 hrs.

Hence, Team 1 will reach B at 9.30, and come to A at 10.00 hrs. After that, they will go to E at 10.30 hrs.

Since Team 4 never passes through stations B, D, or F. Team 4 only can pass through stations A, E, and C.

Hence, the roots of team 4 to reach station E at 11.30 will be (A-E-A-C-A-E) or (A-E-A-E-A-E).

Since team 1 is already traveling to E from A at 10.00 hrs, at that time team 4 can't choose the same route.

Hence, the final route for team 4 to reach E at 11.30 is (A-E-A-C-A-E), and at 12.00 hrs, team 4 will come back to station A.

Hence, the complete route diagram for team 4 is (A-E-A-C-A-E-A)

Teams	9.00	9.30	10.00	10.30	11.00	11.30	12.00
1	A	B	A	E			
2	A	F	E				
3	A	C	D				
4	A	E	A	C	A	E	A

We can see that team 1 is at station E at 10.30 hrs, and they will reach station B at 11.30 hrs, which is only possible when they travel to B via A.

Hence, the complete route diagram for team 1 is (A-B-A-E-A-B-A). It is also known that Teams 1 and 3 are the only ones in station E at 10:30 hrs.

Teams	9.00	9.30	10.00	10.30	11.00	11.30	12.00
1	A	B	A	E	A	B	A
2	A	F	E				
3	A	C	D	E			
4	A	E	A	C	A	E	A

The only possible root for Team 2 at 10.00 hrs is from E to F since they can't choose E to D because Team 3 is already on this route. Since team 3 has to reach A at 12.00. The only possible combination for team 3 is E-D-C-A

Teams	9.00	9.30	10.00	10.30	11.00	11.30	12.00
1	A	B	A	E	A	B	A
2	A	F	E	F			
3	A	C	D	E	D	C	A
4	A	E	A	C	A	E	A

Now the roots for team 2 going back to A is from F at 10.30 hrs (F-A-F-A) or (F-E-F-A).

Hence, the final table is given below:

Teams	9.00	9.30	10.00	10.30	11.00	11.30	12.00
1	A	B	A	E	A	B	A
2	A	F	E	F	A/E	F	A
3	A	C	D	E	D	C	A
4	A	E	A	C	A	E	A

From the table, we can see that at 10.15 hrs, team 3 is travelling from station D to station E.

The correct option is D

[VIDEO SOLUTION](#)

18.D

It is given that the applications are scheduled for processing in twenty 15-minute slots starting at 9:00 am and ending at 2:00 pm. Ten applications are scheduled in each slot.

Hence, the total number of applicants = $(20 \times 10) = 200$. It is also known that 50% of the applications were US applications, and the number of US applications was the same in all the slots. The same was true for the other three categories.

Hence, the number of total number of US applicants = $(200 \times 50\%) = 100$, and the number of US applicants in each slot = $(100/20) = 5$

It is also known that Ira, Vijay, and Nandini were scheduled for Schengen visa processing in that order. They had a 9:15 am slot. Since the number of Schengen applicants was the same in all the slots, it implies the number of Schengen applicants in each slot is at least 3.

Similarly, it is given that Mahira and Osman were scheduled in the 9:30 am slot on that day for visa processing in the Others category, which implies the number of other category applicants in each slot is at least 2. Since the number of total applicants in each slot is 10, this implies the number of Schengen and other applicants in each slot is 3, and 2, respectively. Hence, the number of UK applicants is 0 in each slot.

It is also known that the number of total counters is 10, among which four are dedicated to US applications, and two each for UK applications, Schengen applications, and Others applications. It is given that each US and UK application requires 10 minutes of processing time, and Vijay was called to a counter at 9:25 am. (Who is 5th in the queue). It can only be possible when the processing time of Schengen applications is 12.5 minutes.

US (10 min)				Schengen (12.5 min)		Others (5 min process)	
End Time				End Time		End Time	
C1	C2	C3	C4	C1	C2	C1	C2
9.10	9.10	9.10	9.10	9.12.30	9.12.30	9.05	9.05
9.20	9.25	9.25	9.25	9.25	9.32.30	9.20	9.20
9.30	9.35	9.40	9.40	9.37.30	9.45	9.35	9.35
9.40	9.45	9.50	9.55				
9.55	9.55	10.00	10.05				
10.10	10.10	10.10	10.15				
10.20							

On a particular day, Ira, Vijay, and Nandini were scheduled for Schengen visa processing in that order. They had a 9:15 am slot but entered the VPO at 9:20 am. When they entered the office, exactly six out of the ten counters were either processing applications, or had finished processing one and ready to start processing the next. Hence, at 9.20 am, there are exactly four free counters. Out of these 4, 2 is the UK counter, and the other two are other counters. (Since the US counters and Schengen Counters were either processing applications, or had finished processing one and were ready to start processing the next.)

Nandini's position was sixth in the queue in the Schengen Applications. From the table, we can see that her process will end at 9.45 am.

The correct option is D

[VIDEO SOLUTION](#)

19. C

It is given that

1 burger takes 10 minutes

1 ice cream takes 2 minutes and 3 portions of fries take 5 min by the machine (operator is not required).

Anish or Bani cannot start preparing a new order while a previous order is being prepared.

The first order will be completely done at 10:10.

The second order is two fries and one ice cream, which will be done by 10:15.

The third order is one 1 burger and 1 portion of fries.

The burger will take 10 minutes, by which fries will be ready.

Thus, the third order will be completed by 10:25.

[VIDEO SOLUTION](#)

20. **A**

It is given that

1 burger takes 10 minutes

1 ice cream takes 2 minutes and 3 portions of fries take 5 min by the machine (operator is not required).

The employees are allowed to process multiple orders at a time, but the preference would be to finish orders of clients who placed their orders earlier.

The first order is 1 burger, 1 ice cream and 3 portions of fries.

Anish can start working on the burger and Bani can start working on the ice cream for the first client at 10:00.

The burger will be done at 10:10, ice - cream at 10:02 and fries at 10:05.

The second order is placed at 10:05. (ice cream and fries)

Bani can work on the ice cream for the second client at 10:05 and also put the fries.

The ice cream will be done by 10:07 but the fries will be done by 10:10.

Thus, order will be completed by 10:10.

 VIDEO SOLUTION

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21. **B**

If all the doctors served the patients one after the other, then in 2.5 hrs, Ben will serve 15 patients, Kane will serve 10 patients and Wayne will serve 6 patients.

Ben will earn = $15 \times 100 = \text{Rs } 1500$

Kane will earn = $10 \times 200 = \text{Rs } 2000$

Wayne will earn = $6 \times 300 = \text{Rs } 1800$

So Kane will earn the maximum amount in consultation charges on that day.

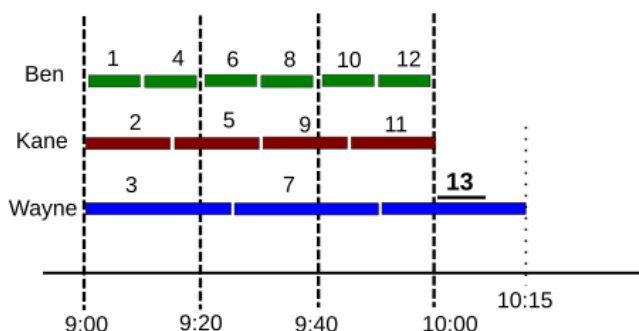
Option B

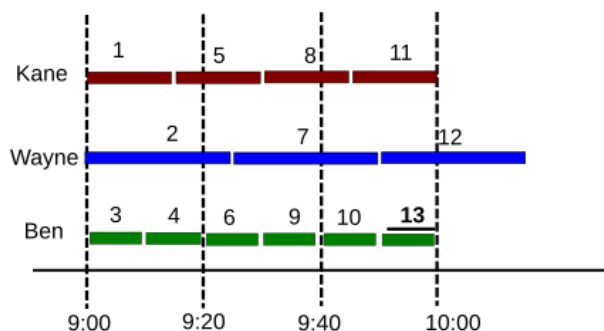
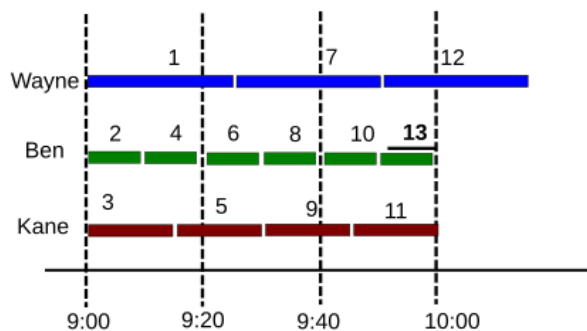
 VIDEO SOLUTION

22. **C**

Mr Singh is 13th in the sequence on all the three days.

The following table will show the sequence for Monday, Wednesday and Friday.





He will stay the longest when the 13th guy is served by Doctor Wayne.

From the table, on Monday he had to wait at the clinic for the maximum duration: till 10:15.

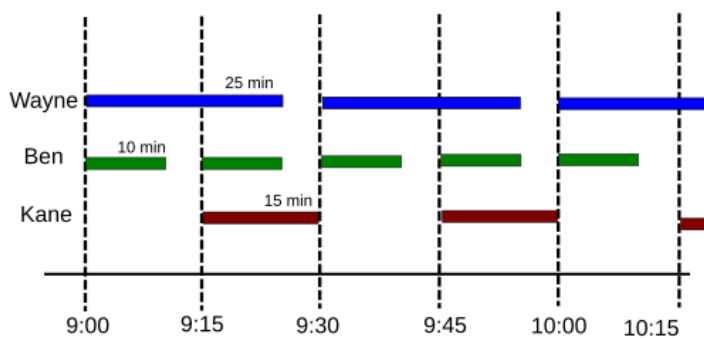
Option C

[VIDEO SOLUTION](#)

23. **D**

On Thursday, the preference order for the patients is Wayne, Ben and Kane.

The first two customers will be served by Wayne and Ben. While Kane will be empty for the first 15 mins. Then he and Ben will serve the next two customers and Wayne will be empty for 5 minutes as shown in the figure below.



As we can see that the cycle will repeat after every 30 mins.

So all three doctors are never simultaneously free.

Option D

[VIDEO SOLUTION](#)

24. **B**

P (2)	Sociology	B, C
Q (1)	Anthropology/ Sociology	Either F or G
R (2)	Economics	<u>A</u> , Either of D/ E/ H
S (1)	Sociology/ Anthropology	G/ F
T (2)	Economics	Any two of the D/ E/ H

As we can clearly say from here that H/D/E are all economics students. Hence Option B is definitely correct

VIDEO SOLUTION

25. D

P (2)	Sociology	B, C
Q (1)	Anthropology/ Sociology	Either F or G
R (2)	Economics	<u>A</u> , Either of D/ E/ H
S (1)	Sociology/ Anthropology	G/ F
T (2)	Economics	Any two of the D/ E/ H

09:00			09:30			10:00			10:30		
Teacher	Student	Subject	Teacher	Student	Subject	Teacher	Student	Subject	Teacher	Student	Subject
T		Economics	T		Economics	R	A	Economics	R		Economics
		Sociology		B	Sociology					F	Anthropology
				G	Sociology						

Now D is scheduled in a slot later than Q. Q can mentor only F or G, but obviously he can't mentor F because F is in the last slot, hence Q will mentor G. In that scenario D has to take the 10:30 slot which means D will be mentored by R and E/H will be mentored by T. So statement 1 and statement 2 both are correct.

VIDEO SOLUTION

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26. C

P (2)	Sociology	B, C
Q (1)	Anthropology/ Sociology	Either F or G
R (2)	Economics	<u>A</u> , Either of D/ E/ H
S (1)	Sociology/ Anthropology	G/ F
T (2)	Economics	Any two of the D/ E/ H

09:00			09:30			10:00			10:30		
Teacher	Student	Subject	Teacher	Student	Subject	Teacher	Student	Subject	Teacher	Student	Subject
T		Economics	T		Economics	R	A	Economics	R		Economics
		Sociology		B	Sociology					F	Anthropology
				G	Sociology						

If E and Q are scheduled in the same slot, they both can be in the last slot. In that case E will be guided by R and both D and H will be guided by T. Moreover if they are in the second slot, E will be guided by T and atleast one of D and H will be guided by T. Hence option (c) is correct

VIDEO SOLUTION

27. B

It is given that none of the streets has more than one team traveling along it in any direction at any point in time (point 1), which implies at 9.00 hrs, all 4 teams have chosen different roots from the starting point.

It is also known that Teams 2 and 3 are the only ones in stations E and D respectively at 10:00 hrs, and Team 1 and Team 4 are the only teams that patrol the street connecting stations A and E.

It is only possible when Team 2 traveled (A-E) via F, and Team 3 reached station D via station C.

It is also known that Teams 1 and 3 are the only ones in Station E at 10:30 hrs, and Team 4 never passes through Stations B, D, or F. Hence, Team 1 must have chosen the (A-B) root at the starting point, and Team 4 has chosen the (A-E) root at 9.00 hrs.

Hence, Team 1 will reach B at 9.30, and come to A at 10.00 hrs. After that, they will go to E at 10.30 hrs.

Since Team 4 never passes through stations B, D, or F. Team 4 only can pass through stations A, E, and C.

Hence, the roots of team 4 to reach station E at 11.30 will be (A-E-A-C-A-E) or (A-E-A-E-A-E).

Since team 1 is already traveling to E from A at 10.00 hrs, at that time team 4 can't choose the same route.

Hence, the final route for team 4 to reach E at 11.30 is (A-E-A-C-A-E), and at 12.00 hrs, team 4 will come back to station A.

Hence, the complete route diagram for team 4 is (A-E-A-C-A-E-A)

Teams	9.00	9.30	10.00	10.30	11.00	11.30	12.00
1	A	B	A	E			
2	A	F	E				
3	A	C	D				
4	A	E	A	C	A	E	A

We can see that team 1 is at station E at 10.30 hrs, and they will reach station B at 11.30 hrs, which is only possible when they travel to B via A.

Hence, the complete route diagram for team 1 is (A-B-A-E-A-B-A). It is also known that Teams 1 and 3 are the only ones in station E at 10:30 hrs.

Teams	9.00	9.30	10.00	10.30	11.00	11.30	12.00
1	A	B	A	E	A	B	A
2	A	F	E				
3	A	C	D	E			
4	A	E	A	C	A	E	A

The only possible root for Team 2 at 10.00 hrs is from E to F since they can't choose E to D because Team 3 is already on this route. Since team 3 has to reach A at 12.00. The only possible combination for team 3 is E-D-C-A

Teams	9.00	9.30	10.00	10.30	11.00	11.30	12.00
1	A	B	A	E	A	B	A
2	A	F	E	F			
3	A	C	D	E	D	C	A
4	A	E	A	C	A	E	A

Now the roots for team 2 going back to A is from F at 10.30 hrs (F-A-F-A) or (F-E-F-A).

Hence, the final table is given below:

Teams	9.00	9.30	10.00	10.30	11.00	11.30	12.00
1	A	B	A	E	A	B	A
2	A	F	E	F	A/E	F	A
3	A	C	D	E	D	C	A
4	A	E	A	C	A	E	A

From the table, we can see that among the options station E is visited the largest number of times.

[VIDEO SOLUTION](#)

28. **A**

It is given that there were a total of 3 rooms and seven candidates. Ganeshan said that he was one among the two candidates allotted to Room 102 whereas Erina said that she is the only person in the room she was allotted to. Therefore, we can say that there were 1 and 4 candidates in either 101 or 103 room. But it is given that Balaram was the third person to enter in the Room 101 therefore we can say that there were 4 candidates in Room 101 and only 1 candidate in room 103.

Room no.	Candidates	Name
101	4	
102	2	
103	1	

Fatima said that three people including Akhil were already in the room that I was allotted to when I entered it. Hence, we can say that Fatima was the last person to enter in 101 and Akhil is the first person who entered in room 101.

Chitra said that she was the last person to enter the room she was allotted to. Hence, we can say that Chitra was allotted room no 102 and she entered after Ganeshan.

Erina was the only person in room no 103.

Room no.	Candidates	Name (In sequential order)
101	4	Akhil, _____, _____, Fatima
102	2	Ganeshan, Chitra
103	1	Erina

Balaram said he was third to enter room no 101. Hence, we can say that Divya was second person who entered in room 101.

Since Chitra and Fatima were already in by 7:40 AM we can say that the candidate who entered at 7:45 am is Erina.

Room no.	Candidates	Name (In sequential order)
101	4	Akhil(7:10), Divya, Balaram, Fatima(7:40)
102	2	Ganeshan, Chitra(7:30)
103	1	Erina(7:45)

From the table we can see that Erina reached the venue at 7:45 am. Hence, option A is the correct answer.

[VIDEO SOLUTION](#)

29. **D**

It is given that there were a total of 3 rooms and seven candidates. Ganeshan said that he was one among the two candidates allotted to Room 102 whereas Erina said that she is the only person in the room she was allotted to. Therefore, we can say that there were 1 and 4 candidates in either 101 or 103 room. But it is given that Balaram was the third person to enter in the Room 101 therefore we can say that there were 4 candidates in Room 101 and only 1 candidate in room 103.

Room no.	Candidates	Name
101	4	
102	2	
103	1	

Fatima said that three people including Akhil were already in the room that I was allotted to when I entered it. Hence, we can say that Fatima was the last person to enter in 101 and Akhil is the first person who entered in room 101.

Chitra said that she was the last person to enter the room she was allotted to. Hence, we can say that Chitra was allotted room no 102 and she entered after Ganeshan.

Erina was the only person in room no 103.

Room no.	Candidates	Name (In sequential order)
101	4	Akhil, _____, _____, Fatima
102	2	Ganeshan, Chitra
103	1	Erina

Balaram said he was third to enter room no 101. Hence, we can say that Divya was second person who entered in room 101.

Since Chitra and Fatima were already in by 7:40 AM we can say that the candidate who entered at 7:45 am is Erina.

Room no.	Candidates	Name (In sequential order)
101	4	Akhil(7:10), Divya, Balaram, Fatima(7:40)
102	2	Ganeshan, Chitra(7:30)
103	1	Erina(7:45)

From the table we can see that Ganeshan is the first person to enter in room 102. Hence, option D is the correct answer.

▶ VIDEO SOLUTION

30. A

It is given that there were a total of 3 rooms and seven candidates. Ganeshan said that he was one among the two candidates allotted to Room 102 whereas Erina said that she is the only person in the room she was allotted to. Therefore, we can say that there were 1 and 4 candidates in either 101 or 103 room. But it is given that Balaram was the third person to enter in the Room 101 therefore we can say that there were 4 candidates in Room 101 and only 1 candidate in room 103.

Room no.	Candidates	Name
101	4	
102	2	
103	1	

Fatima said that three people including Akhil were already in the room that I was allotted to when I entered it. Hence, we can say that Fatima was the last person to enter in 101 and Akhil is the first person who entered in room 101.

Chitra said that she was the last person to enter the room she was allotted to. Hence, we can say that Chitra was allotted room no 102 and she entered after Ganeshan.

Erina was the only person in room no 103.

Room no.	Candidates	Name (In sequential order)
101	4	Akhil, _____, _____, Fatima
102	2	Ganeshan, Chitra
103	1	Erina

Balaram said he was third to enter room no 101. Hence, we can say that Divya was second person who entered in room 101.

Since Chitra and Fatima were already in by 7:40 AM we can say that the candidate who entered at 7:45 am is Erina.

Room no.	Candidates	Name (In sequential order)
101	4	Akhil(7:10), Divya, Balaram, Fatima(7:40)
102	2	Ganeshan, Chitra(7:30)
103	1	Erina(7:45)

From the table we can see that Divya was allotted room no 101. Hence, option A is the correct answer.

▶ VIDEO SOLUTION

CAT Percentile Predictor

31. D

In the information, it is provided that there are four authors in the logistics department, 2 authors in automation, and 2 authors in manufacturing. There are 4 Indian authors, 2 Chinese authors, and 2 Japanese authors.

The total issues publicized were 18 of which the publications done by A, D, G, H are twice the papers written by the other four authors. Hence $A+D+G+H$ has written 12 papers in total and $B+C+E+F$ has written 6 in total. Since an author can write a minimum of 4 papers and a maximum of 3 papers. Each of A, D, G, H must have written 3 papers each.

In statement 3 it was provided that none of the Indian authors were from the Manufacturing area and none of the Japanese or Chinese authors were from the Automation area.

Since none of the Japanese and Chinese authors belonged to automation. Hence the two automation authors must be from India. In statement 1 it was given that F an Indian author is from logistics.

Of the remaining 5 authors 3 from logistics and 2 from manufacturing, in statement 5 it is given that the two Chinese authors are from different areas and hence they must be from manufacturing and logistics.

Of the remaining 3 authors 2 in logistics and 1 in manufacturing, in statement 8 is given that a Japanese author belonged to manufacturing, and hence of the remaining 2 in logistics one of them was from India and the other from Japan.

Of the four Indian authors, 2 belonged to logistics and 2 automation. Of the two authors from Japan, one belonged to manufacturing and one logistics. Of the two Chinese authors, one of them belonged to logistics and the other manufacturing.

Five papers were scheduled in each of the January and April issues, while four were scheduled in each of July and October issues.

Using statement 1, F an Indian author wrote a single paper in logistics published in October.

Using statement 2, A was from Automation and did not have a paper scheduled in October. Since A wrote 3 papers in total he must have written in the other three months and since only Indian authors worked in Automation he must have been from India.

Using statement 5, C, E are Chinese and wrote an equal number of papers. Since $B+C+E+F$ have written a total of 6 papers. The two possibilities are $2+2+1+1$ or $3+1+1+1$. Since it was given that E had papers scheduled in consecutive issues and C did not. So the only possible case is $C = 2, E = 2, F = 1, B = 1$.

Using statement 4, A and H were from different countries and wrote papers in the same months. Hence H must be Japanese and must have written in Jan, April, July.

Author	Country	Area of Interest	Publications	Months of publication
A	India	Automation	3	Jan, April, July
B			1	
C	Chinese		2	
D			3	
E	Chinese		2	
F	India	Logistics	1	October
G			3	
H	Japan		3	Jan, April, July

In statement 6, B, from the Logistics area, had a paper scheduled in the April issue of the journal.

In statement 7, B and G belonged to the same country. None of their papers were scheduled in the same issue of the journal. Since B and G belonged to the same country the only possibility is that both of them belonged to the same country and since they published Journals in different months. G must have published in January, July, and October.

In statement 8, D, a Japanese author from the Manufacturing area, did not have a paper scheduled in the July issue. Hence he must have had the three issues in Jan, April, Oct. The other Japanese author H must have written in logistics.

The fourth Indian author G must have written a paper in Automation.

In January, one more paper needs to be publicized, one in April, 1 in July, and one in October.

In statement 5, E had papers scheduled in consecutive issues of the journal but C did not.

Hence C must have written in Jan and October, and E in April and July.

In statement 9, C and H belonged to different areas. Hence C must be from Manufacturing and E must be from Logistics.

Author	Country	Area of Interest	Publications	Months of publication
A	India	Automation	3	Jan, April, July
B	India	Logistics	1	April
C	China	Manufacturing	2	Jan, October
D	Japan	Manufacturing	3	Jan, April, October
E	China	Logistics	2	April, July.
F	India	Logistics	1	October
G	India	Automation	3	Jan, July, October
H	Japan	Logistics	3	Jan, April, July

There are no authors from Manufacturing in the July issue and hence option D is false.

 VIDEO SOLUTION

32.8

In the information, it is provided that there are four authors in the logistics department, 2 authors in automation, and 2 authors in manufacturing. There are 4 Indian authors, 2 Chinese authors, and 2 Japanese authors.

The total issues publicized were 18 of which the publications done by A, D, G, H are twice the papers written by the other four authors. Hence A+D+G+H has written 12 papers in total and B+C+E+F has written 6 in total. Since an author can write a minimum of 4 papers and a maximum of 3 papers. Each of A, D, G, H must have written 3 papers each.

In statement 3 it was provided that none of the Indian authors were from the Manufacturing area and none of the Japanese or Chinese authors were from the Automation area.

Since none of the Japanese and Chinese authors belonged to automation. Hence the two automation authors must be from India. In statement 1 it was given that F an Indian author is from logistics.

Of the remaining 5 authors 3 from logistics and 2 from manufacturing, in statement 5 it is given that the two Chinese authors are from different areas and hence they must be from manufacturing and logistics.

Of the remaining 3 authors 2 in logistics and 1 in manufacturing, in statement 8 is given that a Japanese author belonged to manufacturing, and hence of the remaining 2 in logistics one of them was from India and the other from Japan.

Of the four Indian authors, 2 belonged to logistics and 2 automation. Of the two authors from Japan, one belonged to manufacturing and one logistics. Of the two Chinese authors, one of them belonged to logistics and the other manufacturing.

Five papers were scheduled in each of the January and April issues, while four were scheduled in each of July and October issues.

Using statement 1, F an Indian author wrote a single paper in logistics published in October.

Using statement 2, A was from Automation and did not have a paper scheduled in October. Since A wrote 3 papers in total he must have written in the other three months and since only Indian authors worked in Automation he must have been from India.

Using statement 5, C, E are Chinese and wrote an equal number of papers. Since B+C+E+F have written a total of 6 papers. The two possibilities are 2+2+1+1 or 3+1+1+1.. Since it was given that E had papers scheduled in consecutive issues and C did not. So the only possible case is C = 2, E = 2, F = 1, B = 1.

Using statement 4, A and H were from different countries and wrote papers in the same months. Hence H must be Japanese and must have written in Jan, April, July.

Author	Country	Area of Interest	Publications	Months of publication
A	India	Automation	3	Jan, April, July
B			1	
C	Chinese		2	
D			3	
E	Chinese		2	
F	India	Logistics	1	October
G			3	
H	Japan		3	Jan, April, July

In statement 6, B, from the Logistics area, had a paper scheduled in the April issue of the journal.

In statement 7, B and G belonged to the same country. None of their papers were scheduled in the same issue of the journal. Since B and G belonged to the same country the only possibility is that both of them belonged to the same country and since they published Journals in different months. G must have published in January, July, and October.

In statement 8, D, a Japanese author from the Manufacturing area, did not have a paper scheduled in the July issue. Hence he must have had the three issues in Jan, April, Oct. The other Japanese author H must have written in logistics.

The fourth Indian author G must have written a paper in Automation.

In January, one more paper needs to be publicized, one in April, 1 in July, and one in October.

In statement 5, E had papers scheduled in consecutive issues of the journal but C did not.

Hence C must have written in Jan and October, and E in April and July.

In statement 9, C and H belonged to different areas. Hence C must be from Manufacturing and E must be from Logistics.

Author	Country	Area of Interest	Publications	Months of publication
A	India	Automation	3	Jan, April, July
B	India	Logistics	1	April
C	China	Manufacturing	2	Jan, October
D	Japan	Manufacturing	3	Jan, April, October
E	China	Logistics	2	April, July.
F	India	Logistics	1	October
G	India	Automation	3	Jan, July, October
H	Japan	Logistics	3	Jan, April, July

Indian authors wrote a total of $3+1+1+3 = 8$ papers

 VIDEO SOLUTION

33. **C**

If all the doctors served the patients one after the other, then in 2.5 hrs, Ben will serve 15 patients, Kane will serve 10 patients and Wayne will serve 6 patients.

A total of 31 patients can be served on a particular day.

 VIDEO SOLUTION

34. **A**

It is given that there were a total of 3 rooms and seven candidates. Ganeshan said that he was one among the two candidates allotted to Room 102 whereas Erina said that she is the only person in the room she was allotted to. Therefore, we can say that there were 1 and 4 candidates in either 101 or 103 room. But it is given that Balaram was the third person to enter in the Room 101 therefore we can say that there were 4 candidates in Room 101 and only 1 candidate in room 103.

Room no.	Candidates	Name
101	4	
102	2	
103	1	

Fatima said that three people including Akhil were already in the room that I was allotted to when I entered it. Hence, we can say that Fatima was the last person to enter in 101 and Akhil is the first person who entered in room 101.

Chitra said that she was the last person to enter the room she was allotted to. Hence, we can say that Chitra was allotted room no 102 and she entered after Ganeshan.

Erina was the only person in room no 103.

Room no.	Candidates	Name (In sequential order)
101	4	Akhil, _____, _____, Fatima
102	2	Ganeshan, Chitra
103	1	Erina

Balaram said he was third to enter room no 101. Hence, we can say that Divya was second person who entered in room 101.

Since Chitra and Fatima were already in by 7:40 AM we can say that the candidate who entered at 7:45 am is Erina.

Room no.	Candidates	Name (In sequential order)
101	4	Akhil(7:10), Divya, Balaram, Fatima(7:40)
102	2	Ganeshan, Chitra(7:30)
103	1	Erina(7:45)

In the question it is given that Ganeshan entered the venue before Divya. Therefore, we can say that Ganesh must have entered with Akhil at 7:10 am. In that case, Divya and Balaram must have entered at 7:15 am and 7:25 am respectively. Hence, option A is the correct answer.

 VIDEO SOLUTION

